RESERVING TIME FOR DADDY: THE SHORT AND LONG-RUN CONSEQUENCES OF FATHERS' QUOTAS

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January 15, 2015

Abstract

To motivate fathers to take parental leave, several nations have reformed their leave programs to increase compensation and reserve some leave for fathers. Though takeup has increased, it is unclear whether fathers are responding to financial incentives or changed constraints or the 'daddy-only' labels. Moreover, although paternity leave may facilitate temporary changes in the division of household labor, little is known about whether these effects persist in later years. This paper examines these issues while studying the Quebec Parental Insurance Program (QPIP), a landmark reform to parental leave that increased benefits for all parents and reserved some weeks for fathers. I exploit plausibly exogenous program variation and data on benefit claims to first explore the short-run effects on leave behavior. Both regression discontinuity and difference-in-difference analyses suggest that QPIP increased fathers' participation rates by 250% and leave duration by 150%. Further, I find novel evidence of an intra-household fly-paper effect stemming from the 'daddy-only' label: benefits stick to fathers even when the constraint does not bind and the 'daddy-only' benefits are essentially fungible with the mothers'. Next, I investigate the long-term consequences on household dynamics by analyzing time-diary data and exploiting variation in exposure to QPIP across provinces, time and the age of one's children. This paper presents the first comprehensive evidence that exposure to paternity leave can have a large and persistent effect on the division of household labor: households exposed to QPIP are markedly less sex specialized in their time allocations, earnings and labor supply.

JEL Codes: J22, J16, J13, J18

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†I would like to thank Francine D. Blau, John M. Abowd, Michael F. Lovenheim, Lawrence M. Kahn, Kelly Musick, Richard Mansfield, Dennis Becker and participants at the American Economic Association Meetings (January 2014), Labor and Employment Relations Meetings (May 2013), Society of Labor Economists Annual Meeting (May 2014), and the COSME-FEDEA Workshop on Gender Economics (May 2013) for helpful comments. In addition, I gratefully acknowledge financial support for this project from the L.R. Wilson Award, Benjamin Miller Award, Mario Einaudi Center for International Studies, Center for the Study of Inequality, Cornell Population Center and Cornell University Graduate School. This analysis is based on Statistics Canada's Employment Insurance Coverage Survey (2002-2010) and General Social Survey (2005, 2010) restricted-access master files. All computations, use and interpretation of these data are entirely that of the author.

1 Introduction

Despite a dramatic reduction in the gender gap in labor force participation and wages, a large and persistent gap remains in the realm of care work. Mothers are much more likely than fathers to take parental leave in the first months of a child's life, which has the potential to hurt mothers' careers.[1] Consistent with this pattern, a cross-country analysis by the OECD reports that the length of paid parental leave available is correlated with a higher pay differential by gender (OECD, 2012). Mothers also perform considerably more unpaid work in the home than do fathers (Hochschild and Machung, 1989; Blair and Lichter, 1991; Bianchi, 2011, 2012). In turn, this disproportionate amount of housework done by women, particularly time-inflexible and routine work, has been shown to contribute to the gender pay gap (Hersch and Stratton, 2002). Becker (1981) used the idea of comparative advantage to argue that the traditional division of household labor may be efficient due to women's lower market wages and biological advantages in care-giving such as the ability to breastfeed. However, recent decades have witnessed considerable growth in women's wages and the advent of technology to minimize biological differences, without a corresponding reduction of the same magnitude in household sex specialization. These patterns suggest that sticky social norms about gender roles may be perpetuating higher levels of specialization than necessarily efficient for the household.[2] Consequently, men's participation in parental leave and care work has attracted growing public attention, and several nations have enacted policies to promote fathers' participation in paid parental leave programs.

These policies are motivated by the idea that active fathering not only benefits child development (Tamis-LeMonda and Cabrera, 2002; Lamb, 2010; El Nokali et al., 2010) and improves men's satisfaction with contact with their children (Haas and Hwang, 2008), but also may promote gender equality. Fathers' participation in parental leave strengthens women's labor market positions by not only enabling mothers to return to work earlier but also eroding the rationale behind statistical discrimination against women. Further, paternity leave makes the initial parenting experience more similar across genders, which can have long-term consequences for household behavior. In the absence of such schemes, mothers often leave the labor market for varying amounts of time and increase investments in childcare and housework, while fathers maintain or even strengthen their ties to the workforce (Sanchez and Thomson, 1997), leading to differential investments in non-market and market skills. This establishes a gendered division of household labor, which

[1] Several studies note that the provision of extended maternity leave delays women's return to work (Schönberg and Ludsteck, 2007; Lalive and Zweimller, 2009) and lowers the probability of upward occupational moves (Evertsson and Duvander, 2011). In addition, Blau and Kahn (2013) find that generous parental leave policies seem to encourage women's part-time work and employment in lower level positions.

[2] Sex specialization may be undesirable for many reasons beyond just inefficiency, as argued by Blau et al. (2014). Since it reduces a woman's wages and increases her marriage-specific investments, female specialization in care work lowers her bargaining power. The costs of interdependence are also disproportionately borne by women, as in the case of the displaced homemaker who upon being widowed or divorced must support herself from a weakened market position.

persists after the leave period ends. So it can be argued that if mothers and fathers experience more similar transitions to parenthood, it might reduce sex specialization in the long run. Keeping these arguments in favor of paternity leave in mind, we are faced with two broad questions. First, what kinds of incentives are successful in getting parents to share leave more equally and what are the mechanisms behind that success? Second, if a policy succeeds in getting fathers to take leave, does this lead to more egalitarian households in the long run, or are social norms so sticky that parents revert to traditional gender roles once the leave period ends?

In this paper I explore these questions while providing the first comprehensive causal analysis of the short- and long- term consequences of a policy aiming to promote paternity leave. I study the Quebec Parental Insurance Program (QPIP), a system of parental leave benefits introduced in Quebec in 2006 that explicitly sought to boost fathers' participation in paid parental leave. From 2001 to 2005, eligible parents in all Canadian provinces could claim parental leave benefits from the government through the Employment Insurance (EI) Program. In 2006, Quebec left the EI system and established the Regime Quebecois D'assurance Parentale or the Quebec Parental Insurance Plan. This new scheme lowered eligibility criteria, increased income replacement, and established a 5-week 'daddy quota' of leave for fathers (Doucet et al., 2010). Due to QPIP's 'daddy quota', Quebec is the only province in Canada in which fathers enjoy an individual and non-transferable right to parental leave. Prior to the reform, fathers only had access to 'shared' parental leave with their spouses, and leave-takers were compensated with a little over half their wages up to a strict cap so that household incomes were hit hard when fathers took leave. Consequently, fathers' leave participation never exceeded 22% prior to QPIP. Notably, the majority of families did not exhaust their total amount of leave prior to the reform, such that families were leaving benefits 'on the table' even as fathers declined to participate.

This study makes several important contributions to the literature. First, this is the first study to explore the mechanisms behind why daddy quotas may be effective. I am able to not only analyze the extent of QPIP's impact on leave behavior but also detangle the effects of different reform features, e.g., the effect of improving benefits versus that of assigning a binding quota versus that of a 'daddy-only' label. Consequently, this paper is the first to explore whether fathers respond to 'daddy-only' labels, which would be suggestive of an intra-household flypaper effect (IHFE) whereby leave 'sticks' to fathers even if the quota does not change a binding constraint. Second, this paper offers the first comprehensive examination of the long-term causal effects of paternity leave. By simultaneously investigating multiple outcomes related to gender equality, such as the division of household labor, patterns of housework and spouses' incomes, I offer evidence of the impact on overall household dynamics. Third, since only the province of Quebec deviated from the national policy,

this study is also unique in utilizing regional variation in policy rather than a nationwide change in policy. This results in improved study design: by using other Canadian provinces where policy did not change as a natural control group, I can provide causal estimates that are robust to various trends. Fourth, this is the first causal study to leverage data from time-diaries, which are increasingly considered the gold-standard for gathering information about contributions to non-market work (Sevilla, 2014). Consequently, I am able to explore more accurate and detailed measures of parent's behavior than prior studies on this subject have been able to. Lastly, this is the first study of how this Canadian policy episode affected gendered patterns in leave behavior and market and non-market production. This is interesting because Canada offers a political and social context that is quite different from the previously-studied Scandinavian countries, since the latter have some of the most generous welfare provisions and family-friendly policies worldwide. In sum, this study contributes to the literature by utilizing better data and improved methodology in a new context, as well as examining the causal channels behind the daddy quota's effect and the bigger picture of sex specialization across the breadth of parental responsibilities.

To investigate the short-run effects of QPIP on parents' leave behavior, I analyze data on benefit claims from the 2002-2010 rounds of the Employment Insurance Coverage Survey (EICS). I use a sharp regression discontinuity design to identify the local mean impact of QPIP at the point when it was introduced, and a difference-in-differences approach to estimate the average treatment effect of QPIP since it has been introduced. Both sets of results show that QPIP was very effective in achieving its goal of boosting fathers' involvement. The introduction of QPIP was associated with an increase in fathers' claim rates of 53 percentage points and an increase in fathers' leave duration of 3 weeks. There is some evidence that QPIP also increased mothers' participation, but the effect is much smaller than that for fathers, in both absolute and relative magnitude. My results also suggest that fathers responded to not only the higher benefits but also the 'daddy-only' label in itself. Even though the quota did not alter a binding constraint for many families, reserving some weeks as 'daddy-only' shifted the distribution of leave towards fathers. That is, under the new program fathers took paid leave that they would have had available even prior to QPIP. This paper thus provides novel evidence of an intra-household flypaper effect in parental leave - an odd and important finding in terms of policy design, as it suggests that labeling may play an important role in influencing program participation.

Next, I investigate whether the increase in fathers' leave-taking under QPIP had an impact on household division of labor 1-3 years after the leave period ended. I use time-diary data from the 2005 and 2010 rounds of the General Social Survey, exploiting variation in exposure to QPIP across time, provinces and children's ages. I find strong evidence that by altering the initial experience of parental leave, QPIP had a

large and persistent impact on gender dynamics within households. Exposure to QPIP moved households towards a dual-earner, dual-caregiver model wherein fathers and mothers contribute more equally to home and market production. I find strong evidence that exposed mothers spent more time in paid work, spent more time physically at the workplace and were more likely to be full-time employed, compared to their counterparts who were not exposed. Consequently, exposed mothers experienced a large increase in annual personal income. There is also evidence that suggests exposed fathers spend less time in paid work and at their workplace, and also earned less annual income compared to fathers who were not exposed. In the realm of non-market production, I find that exposure to QPIP increased both parents' contributions - although exposed fathers increased their time by more than exposed mothers. Specifically, exposed fathers spend more time in housework per day, while exposed mothers decreased their housework and spent more time in childcare instead. Moreover, exposed fathers spent more time physically at home while exposed mothers spent less time in the home. Overall, a clear pattern of reduced sex specialization emerges among exposed households.

Taken together, these results shed light on how leave schemes can be designed to induce fathers to participate, and they suggest that small changes in the initial parenting experience can have lasting effects on parents' behavior. In particular, they indicate that people respond to labels, and that behavior learned at the beginning tends to stick in later years. More broadly, my findings highlight that there need not be a trade-off between gender equality and parental time with children . By encouraging fathers to participate in parental leave we can distribute household responsibilities more equally and actually increase total time investments in children.

The paper proceeds as follows. Section 2 provides background on the reform. Section 3 discusses the existing literature. Section 4 discusses the data, methodology, results and robustness checks for my analysis of short-run effects, and Section 5 does the same for my analysis of long-term effects. Section 6 concludes.

2 Background

2.1 Parental Leave Programs in Canada and the QPIP Reform

In every Canadian province, at least a year of job-protected parental leave is available to every parent who has worked 52 weeks or more with their current employer.[3] Further, every parent who meets certain eligibility criteria can claim benefits, converting some of this leave into paid leave. The Employment Insurance (EI) Program, which all Canadian provinces used until 2005, offers maternity benefits that mothers can take

[3]The length of job-protected leave does not change over the period of my analysis.

in the weeks immediately succeeding the birth as well as parental benefits that mothers and fathers must decide how to share between them. Most provinces continue to subscribe to the EI Program, with the notable exception of Quebec. On the 1st of January 2006, Quebec introduced the Quebec Parental Insurance Plan (QPIP), to which employees now contribute and claim benefits from instead of the EI system. It should be noted that both the EI and QPIP program are financed through payroll taxes. The details of the EI program, currently offered to residents of other provinces, and the QPIP program, currently offered to residents of Quebec, are shown in Table I.

QPIP's features were designed to offer an improvement over the older EI system by easing some of the barriers that parents face to taking leave, namely, inflexibility, ineligibility, financial feasibility, and gendered attitudes. First, the new system was designed to be more flexible, offering parents a choice between the Basic Plan or a Special plan that offers higher benefits for a shorter duration, thereby letting parents select the combination of benefit amount and duration that best suited their needs. Second, the reform lowered the eligibility criteria in order to improve coverage and ease access to benefits. The EI system requires a claimant to have worked 600 hours of insurable employment. This makes it difficult for workers from seasonal, temporary, part-time or otherwise non-standard employment, who tend disproportionately to be low-income mothers, to qualify for benefits. In comparison, QPIP uses an earnings-based threshold that is easier to meet, such that any parent who has at least 2000CAD of insurable earnings can qualify. Third, QPIP offers more generous compensation for foregone income. By both increasing the maximum replacement rate (from 55% to 70%) and raising the ceiling of maximum insurable earnings on which one can claim (from 39,000CAD to 57,000CAD in 2006), QPIP ensures that a greater portion of foregone wages can be recovered via benefits while on parental leave.

QPIP also introduced the nation's first of its kind 'daddy quota', whereby 5 weeks of leave (or 3 weeks under the Special Plan) were set aside for the father and could not be transferred to the mother. This important feature of the reform stands in stark contrast to the EI Program, where fathers enjoy no individual right to paternity leave and may only access benefits through shared parental leave. More generally, QPIP changed the distribution of benefits within the household. To begin with, QPIP abolished the 2-week waiting period that EI claimants are subject to. Also, the amount of gender-neutral leave to be shared between parents was reduced and some weeks were reallocated to individual non-transferable leave for each parent. The net result was that mothers retained access to the same amount of potential leave as before (50 weeks) but a larger share now came through maternity leave rather than shared parental leave. Fathers gained access to more leave than they had earlier: 37 potential weeks under QPIP (5 of which are 'daddy-only') versus 35 weeks under the EI Program. In total, QPIP increased the amount of leave available to a family

from 50 weeks to 55 weeks, such that total leave increased by the amount equivalent to the 'daddy-only' weeks.[4]

2.2 Expected Impact of QPIP on Parents' Leave Behavior

QPIP's choice of two programs and easier eligibility criteria are not expected to impact fathers significantly. Since the majority of fathers are full-time, full-year workers, they face no difficulty qualifying for benefits under either the EI or the QPIP scheme. Further, since under QPIP the whole family had to act on either the Basic Plan or the Special Plan once the choice was made, few families selected the Special Plan, which limited their duration in return for higher compensation. Therefore, the two changes most likely to influence the decision for fathers to take leave were that of improved benefits and the daddy quota.

First, I consider how the representative parent might respond to increased income replacement. By making benefits more generous, QPIP reduced the opportunity cost of taking leave, i.e. the difference between foregone wages and benefits. Assuming leave is a normal good, the price reduction should result in a positive income and substitution effect, leading to an unambiguous increase in the amount of leave consumed. However, it should be noted that benefits increased for both mothers and fathers, so there is no reason to believe fathers should respond more strongly than mothers to the improved benefits. To the contrary, given the evidence showing that women have more elastic labor supplies than men (Juhn and Murphy, 1997; Blau and Kahn, 2007), if anything we would expect women to respond more strongly to the improved benefits. Further, since benefits are capped at a certain threshold, parents with lower incomes experience a larger marginal reduction in price as they face a smaller wage-benefit differential. Therefore, we should also expect parents from low-income households to respond more strongly than those from high-income households.

Second, I consider the reservation of the daddy quota. A daddy quota could make the difference between a father participating or not participating if, absent the quota, his wife consumed the total amount of leave allocated to the family. In that case, the addition of 5 daddy-only weeks would mean that in order for a family to continue exhausting total family leave, it would be necessary for the father to participate. However, Quebec's reform presents an interesting case because prior to the quota, most families did not use all of their leave. Figure I presents the probability distribution function of the distribution of maternal leave duration in Quebec in the period before the reform. Although there was bunching at the cap of 52 weeks, i.e. 12 months, a significant portion of mothers were not consuming all the paid leave available to the household.

[4] It should be noted that under either the EI or QPIP program, parents can take leave simultaneously so the mother does not have to resume work in order for the father to participate in parental leave.

Figure II shows the cumulative distribution function of mothers' leave duration in Quebec and the Other Provinces in the pre-reform period (2002-2005). Even when the father did not take leave (the majority of families), over 60% of mothers reported taking 11 months or fewer of leave, leaving at least one full month of benefits unused. Furthermore, since the EICS survey asks mothers about all leave taken and not specifically paid parental leave, their answers include the 2 weeks of unpaid 'waiting period' under the EI program, and may even include other kinds of paid leave such as vacation or sick days. For example, a mother who reports taking 11 months of leave is at most taking 10.5 months of paid parental leave, a full 1.5 months less than the maximum the family is entitled to. This implies that for these families, a considerable amount of leave was always available to fathers even though they chose not to use it. Therefore, for the majority of families who were not consuming at the cap pre-reform, the newly imposed constraint of the daddy quota should not have been binding. Accordingly, any increase in total family leave under QPIP should have been considered an ordinary extension of family leave since the additional weeks were essentially fungible between parents.

Given that the constraint does not bind, I investigate whether the daddy quota could alter the parents' maximization problem in a different way. To do so, I consider the existence of a non-monetary cost of leave. In addition to the opportunity cost of taking leave (wages net of benefits), parents may face a non-monetary 'stigma' cost that causes them to discount benefit income compared to wage income. This 'stigma' cost could encompass any number of things, for example, personal distaste to taking leave, peer pressure or workplace hostility to leave-takers. Further, this cost may differ across individuals, for example, stigma may be higher for fathers than mothers, or for fathers working in blue-collar environments versus white-collar environments. Differences in stigma may therefore contribute to the differences in leave participation rates across genders and income or education groups.

One mechanism through which a daddy quota can have an impact even when the constraint does not bind is if the 'daddy-only' label attached to the quota reduces this stigma cost. The daddy-only label establishes a father's individual right to leave, removes the need to negotiate with his wife, and improves his bargaining position with employers and co-workers who may be more sympathetic to him using leave specifically designated for him. Moreover, the quota sends a clear public message that promotes fathers' involvement, which may reduce social stigma against taking leave and possibly even introduce stigma against those who do not utilize this generous opportunity to spend time with their children. The idea that fathers respond to reduced stigma would also be consistent with the finding of Dahl et al. (2014) that fathers are more likely to take parental leave if their brothers or coworkers have done so. Therefore, under QPIP fathers may have experienced a reduction in not only the opportunity cost of taking leave but also the stigma cost. Mothers, however, only experienced the former. This difference might lead to fathers responding more

strongly to the reform than mothers.

We thus can consider two alternate hypotheses:

(i) H_0: No stigma cost exists

In the absence of any stigma cost, parents should respond only to the opportunity cost of taking leave. Since QPIP's improved benefits lowered the opportunity cost for both males and females, we should see leave consumption increase for both parents. Given that women have higher elasticities of labor supply (Juhn and Murphy, 1997; Blau and Kahn, 2007) and tend to be the lower-earning spouse (thereby experiencing a greater marginal increase in benefits), we expect mothers to increase their leave consumption by at least as much as fathers.

(ii) H_1: A stigma cost exists

If a stigma cost exists, which is higher for fathers than mothers and which QPIP's 'daddy-only' label reduces, then fathers may increase their leave consumption by more than mothers. That is, if we observe an intra-household flypaper effect, whereby benefits stick to the fathers even though the additional weeks are fungible for most families, this would be evidence consistent with the existence of a stigma cost.

2.3 Expected Long-Run Impact of Paternity Leave on Household Behavior

Paternity leave policies promote gender equality by intervening at a crucial time for renegotiating household work (Hook, 2010) and facilitating a re-allocation of parents' resources across professional and domestic spheres. They make fathers available for time-inflexible housework and childcare, enabling mothers to return to the workforce sooner and invest in their careers. Thus, reduced sex specialization during the period of paternity leave can be explained both by fathers' increased time availability for non-market work and mothers' increased bargaining power.[5] A main objective of this study, however, is to explore the *long-term* causal effects of paternity leave on the household division of labor, which no previous study has been able to establish. I seek to answer the following question: do the effects of paternity leave on sex specialization persist after the leave period, or does the household revert to traditional gender roles afterward? There are several channels through which paternity leave may lead to a *permanent* reduction in sex specialization, as I argue below.

The first explanation builds on the theory of Becker (1981), in which a household uses productivity differentials that may differ across genders to determine an efficient allocation of resources. Since men

[5] If bargaining power is proportional to an individual's contribution to the household income (as proposed by e.g. (Lundberg and Pollak, 1996)), fathers on leave have reduced earnings and thereby diminished ability to bargain away from doing unpleasant domestic chores, while their wives who have returned to work would have higher earnings and therefore bargaining power. This argument relies on the parental leave being compensated at less than 100% of usual earnings, which is the case in Canada and in many nations offering extensive paid leave to fathers.

earn higher market wages on average and women have some biological advantages in childcare, the theory of comparative advantage suggests that men allocate more time to market work while women take on more domestic responsibilities because it is efficient for the household. However, being on leave increases fathers' time in childcare and housework, especially time-inflexible tasks, in which they gain experience and competence. Fathers on leave undergo on-the-job training, which increases their domestic productivity and reduces differentials between their returns to non-market and market work. If fathers are penalized by employers for taking leave through lower wages or fewer promotions, this further reduces their productivity differential by lowering returns to market work.[6] Women whose husbands take paternity leave can return to work earlier and enjoy greater job continuity, increasing their returns to market work. As the ratio of the returns to market versus non-market work for mothers and fathers converge, this should lead to a within-family time allocation that is less sex-specialized, whereby fathers contribute more to unpaid work and mothers contribute more to market work.

Another mechanism through which paternity leave may influence behavior in the long-term is that of habit persistence in preferences. Individuals may have utility over different kinds of work that is non-separable over time. Under such a model, lifetime utility would take the form of $U(C) = \sum_{t=0}^{T} u(c_t - \alpha c_{t-1})$, where $u(.)$ is a concave utility function, c_t is consumption in period t and α denotes the intensity of habit formation. Due to the concavity of the utility function $u(.)$, we would then have that $u'(c_t) \leq 0$ and $u'(c_{t-1}) \geq 0$, that is, that marginal utility is decreasing in current consumption as per usual but is increasing in past consumption. Thus, an increase in current consumption, c_t, will lower the marginal utility of consumption in the current period but increase marginal utility in the next period. For example, paternity leave could increase a father's initial consumption of childcare, which increases the marginal utility of childcare in the next period, such that he demands more contact with his child even after the leave period ends. The same logic would hold if we modeled either childcare or housework as a 'bad' rather than a 'good'. For example, paternity leave may increase a father's participation in cooking, which lowers his marginal dis-utility from the task in future periods.

Paternity leave also may create a pattern of household behavior during the period of leave that is costly to reverse later. One potential cost of changing behavior after the leave period is that of learning. Parents who take leave simultaneously may divide up non-market tasks and each invest in task-specific human capital. After the leave period ends, it becomes costly for either parent to learn how to perform the other's designated task and to avoid this cost they may continue to share chores as they did while they were on

[6] Over time, if social norms and expectations about gender roles change, we may also see men and women make different decisions about human capital investments as well as occupation and industry choices, which would have an impact on their returns to market work.

leave. In addition, there may be utility costs associated with reversion. For example, the wives of men who take leave may enjoy the experience of committing to their careers while being supported by a helpful spouse at home, and they may perceive dis-utility from returning to traditional gender roles where their career is subordinate to their spouse's.[7]

Paternity leave should also limit the possibility of strategic shirking, since fathers cannot credibly claim to be incompetent in certain childcare and housework tasks any longer. Lastly, the public message promoting active fathering behind a daddy quota, as well as the actual experience of taking paternity leave, could influence the identity of fathers and their spouses (Akerlof and Kranton, 2010).

3 Prior Literature

3.1 Previous Research on Leave Policies and Fathers' Participation

Despite the considerable evidence that fathers' involvement in childcare is positively associated with children's social, emotional, physical, and cognitive development (Allen and Daly (2007) provide a useful summary), the leave participation rates of fathers worldwide remain much lower than those of mothers. Since the father is often the higher-earning parent, financial compensation plays a significant role in their decision to take leave. Studies have shown that loss of earnings is an important factor in fathers' decisions to not take parental leave (Zhelyazkova, 2013). It is also common for fathers to cite workplace attitudes as an obstacle to utilizing leave even when they are entitled to it, out of fear it could damage their careers (Bygren and Duvander, 2006). Social and psychological factors also may play a role: it is possible that men have a lower taste for childcare, that social constructs push men to see themselves as the primary breadwinner who must prioritize paid work, or that they are rarely exposed to role models in the form of men who care for infants.

Several studies have exploited cross-country variation in policies to determine how easing these barriers can improve fathers' leave-taking. Fathers' leave take-up tends to be higher in countries with generous compensation rates (Moss and O'Brien, 2006) and is especially low in countries like the United States where leave is unpaid (Han et al., 2007). O'Brien (2009) compares 24 countries and finds fathers' use of statutory leave is greatest when high income replacement (fifty percent or more of earnings) is combined with extended duration (more than fourteen days). It also matters whether fathers' access to leave is derived via a family right or an individual right. Several cross-country comparisons have shown that fathers are more likely to utilize leave when a daddy quota is in place (Bruning and Plantenga, 1999; O'Brien, 2009; Haas and

[7] In that case, under a unitary model of maximization reverting to the traditional division of labor would be sub-optimal for the household. Alternatively, under a non-cooperative model, since mothers benefit from the non-traditional division of labor they can use their improved bargaining power to enforce continuation of this behavior even after the leave period ends.

Rostgaard, 2011). However, while these findings provide suggestive associations between different kinds of leave policies and fathers' behavior, they suffer from endogeneity issues since the assignment of each country to a specific policy regime is non-random, e.g., a country may offer high income replacement precisely because parents are highly motivated or concerned about parental leave.

More recently a few studies have exploited natural experiments, where leave policy was changed suddenly, to identify causal effects by comparing births just before and just after the reform. Dahl et al. (2014) report that the introduction of a daddy quota in Norway had an impact on fathers' takeup of 32 percentage points. Duvander and Johansson (2012) and Ekberg et al. (2013) study Sweden and find a strong effect on parental leave use resulting from the reservation of the first 'daddy month'.[8] These studies present causal estimates of the impact of daddy quotas and provide some evidence of the success of such schemes. However, the specific nature of these reforms present limitations on the ways in which we can interpret the program effects. In the case of Sweden, the daddy quota did not represent the addition of a new month of leave, but instead a transfer from total family leave to 'daddy-only' leave. Thus, if the mother had previously exhausted the total leave, the quota now made it necessary for the father to participate to simply maintain the status quo amount of family leave. In Norway, the introduction of the quota did not decrease mothers' potential leave, but since most mothers took the entire amount of family leave prior to the reform (Dahl et al. (2014), pp.2), the family could only use the additional leave if the father utilized his month. Therefore, in the case of both Norway and Sweden, the introduction of the quota altered a binding constraint. We therefore cannot be sure about the mechanisms behind the reforms' success, i.e., were fathers responding to their individual right and the 'daddy-only' label or were families simply trying to maximize leave, which made it necessary for fathers to participate?

3.2 Previous Research on the Long-Run Effects of Paternity Leave

Much of the extant research on the long-term effects of paternity leave has examined variation in actual leave-taking among fathers or cross-country variation in leave policies. Such studies have found that fathers who take leave are more involved in childcare (Haas, 1990; Brandth and Kvande, 1998; Haas and Hwang, 1999; Tanaka and Waldfogel, 2007; Nepomnyaschy and Waldfogel, 2007) and that the average father's time in childcare is higher in countries with generous paternity leave policies (Fuwa and Cohen, 2007; Sullivan et al., 2009; del Carmen Huerta et al., 2013; Boll et al., 2014).[9] Moreover, paternity leave is correlated with

[8] Duvander and Johansson (2012) detect a smaller but clear effect resulting from the second daddy month, and no effect from the 'gender equality bonus' which rewards couples for sharing leave equally.

[9] Although studies find no evidence of an association between paternity leave and fathers' average time in housework, some do find evidence consistent with increased male participation in time-inflexible and typically-female housework (Brandth and Kvande, 1998; Hook, 2006, 2010; Estes, 2011)

shorter work hours for fathers (Haas and Hwang, 1999; Duvander et al., 2010), and shorter career breaks, longer work hours and improved labor market positions for mothers (Brandth and Kvande, 1998; Pylkkänen and Smith, 2003). Taken together, these cross-sectional studies suggest that paternity leave is correlated with a less traditional division of labor within the household. However, these associations are vulnerable to endogeneity issues. Cross-country studies may be biased upwards by the omission of country-level variables such as institutional or normative contexts. Similarly, studies using cross-sectional variation in actual leave-taking among fathers cannot control for unobserved heterogeneity in preferences, beliefs, motivation, and workplace constraints. Thus, their findings can only be interpreted as informative associations rather than as causal estimates.

More recently a few studies have sought to identify causal effects of paternity leave by comparing the behavior of parents before and after a change in policy that led to a sudden increase in fathers' leave-taking, thus exploiting exogenous variation in leave-experience. Interestingly, these studies of 'natural experiments' do not confirm the results from cross-sectional research, and consequently highlight the importance of a robust study design. First, several studies fail to detect a significant causal impact of paternity leave on the distribution of childcare between parents (Kluve and Tamm, 2009; Rieck, 2012; Ekberg et al., 2013; Ugreninov, 2013). One study did report that paternity leave leads to more equal sharing of housework, but could only detect a significant effect for the chore of laundry (Kotsadam and Finseraas, 2012). Second, these studies are not consistent in their findings on the causal effect on parents' labor market outcomes. While some studies report that paternity leave reduces fathers' earnings (Johansson, 2010; Rege and Solli, 2013), others find no impact on fathers' earnings or work hours (Cools et al., 2011). Similarly, some studies find no causal effect on mothers' labor supply or earnings (Rege and Solli, 2013; Kotsadam et al., 2011), while others report that paternity leave leads to higher or lower maternal earnings (Johansson (2010) and Cools et al. (2011) respectively). Thus, the results from these quasi-experimental studies are not conclusive, but do confirm the suspicion that the relationship between paternity leave and parental behavior may not be as straightforward as the cross-sectional evidence suggests.

The causal literature also has several important shortcomings that may explain the lack of a conclusive result. First, previous studies focused on one or at most two dimensions of parental responsibility, and are therefore unable to identify substitutions between tasks. Some studies only considered outcomes for one parent (e.g. Kotsadam et al. (2011); Ugreninov (2013)) and so cannot capture the fact that mothers' and fathers' time may be complements or substitutes in household production. Second, other causal studies have used measures of parents' non-market contributions that I argue to be deficient. Some studies explore narrow measures of parental involvement, e.g., Ekberg et al. (2013) and Ugreninov (2013) use the share

of leave taken to care for sick children as their measure of a parent's household work. Other studies use broader measures, but rely on data vulnerable to measurement error and reporting bias. For example, Kluve and Tamm (2009) ask parents 1.5 years after the birth to report the proportion of childcare performed by them during the first year of the child's life, even though respondents may incorrectly remember their contributions. Other studies, such as Kotsadam and Finseraas (2012), use surveys that ask the respondent, for example, "*Who does the chore of laundry in your household? - you? -your partner? -you share equally?*". The limited range of possible answers means these measures lack precision. More importantly, since these questions explicitly hint at evaluating gender relations these data are susceptible to response bias, wherein respondents purposefully understate or exaggerate their behavior to align with cultural norms about gender equality rather than reporting their true actions. Lastly, other quasi-experimental studies have exploited nation-wide changes in policy to compare fathers who experienced a birth before a reform to those who experienced a birth after. Analyses using only one period of observation (e.g. Kotsadam and Finseraas (2012)) thus necessarily compare fathers of older children to those of younger children, whose behaviors may differ inherently - and further, the differences may simply reflect trends in parent's behavior. Studies with multiple periods of observation produce difference-in-difference estimates by comparing parents' behavior across children's ages and and time (e.g. Rege and Solli (2013)), but it can be argued that parents of older children are not an ideal comparison group due to the strong identifying assumption of parallel trends in parents' behavior across children's ages.

4 Short-Run Effects on Parents' Leave Behavior

4.1 Data on Benefit Claims

To analyze the immediate impact of QPIP on parents' leave behavior, I use data on benefit claims collected through the Employment Insurance Coverage Survey (EICS) (Statistics Canada, 2002-2010). The target population for this annual survey is a subset of the target population for the Labor Force Survey, and comprises individuals who, given their recent status in the labor market, could potentially be eligible for employment insurance. Mothers of infants less than one year old, who I will focus on in this study, fall into this last category, since they could potentially be eligible for benefits via maternity or parental leave. The EICS is conducted annually, and I focus on mothers in a nine-year window framing the QPIP reform, from 2002 to 2010. Specifically, I use data from 2002-2005 as the pre-reform period (roughly 42% of the

observations), and 2006-2010 as the post-reform period.[10] It should be noted that I use restricted-access versions of this data which can only be accessed on-site at a Statistics Canada Remote Data Center, as the Public Use Microdata do not have detailed information on month of birth, fathers' leave duration and household income.

The primary sample comprises 8,907 observations of mothers aged 18-40 who have a child under one year old and identify as part of a married or cohabitating couple.[11] Approximately one-fifth of the observations are from Quebec. The rest of the observations come from the control group, which comprises the five largest other provinces, i.e. Ontario, Alberta, British Columbia, Atlantic Region, and Manitoba and Saskatchewan, where the EI system remained in place over the entire period of the analysis.

The outcomes regarding leave participation are measured by indicators taking value 1 if the respondent (or her spouse) has claimed or plans to claim maternity/parental/paternity benefits through the EI or QPIP system. Parents' leave duration is measured by mothers' reports of total weeks of actual or planned leave taken by her and her spouse. My measures of leave duration are not conditional on participation and so include zeros, thus offering a summary measure that takes into account both changes in participation and changes in duration conditional on participation.[12] There are two important things to note about the measures of mothers' leave duration. First, mothers who are still on leave at the time of survey offer responses about planned leave duration while mothers who have returned to work report their completed leave duration. There is therefore concern that mothers may report planned duration that is either shorter or longer than the actual length of leave the parent ends up taking. However, since the EICS only covers mothers who have an infant under a year old, limiting our sample to mothers who have already returned to work would lead to the systematic over-representation of mothers who took shorter leaves, skewing the distribution of leave durations to the left. Consequently, I treat duration of leave to be length of completed leave for those who have returned, and length of planned leave for mothers still on leave. Second, the survey asks new mothers about the duration of all leave (not specifically paid parental leave) taken by them, and could capture unpaid leave or paid sick or vacation leave mothers take in lieu of paid parental leave. However, given the generous paid parental leave available and lack of stigma to maternal leave-taking, this

[10] There were nation-wide reforms to both job-protected and paid parental leave in late 2000, and Quebec also extended its publicly subsidized childcare to children aged 0 to 1 in 2001. This motivates me to exclude data from survey years prior to 2002 as well as any observations in survey year 2002 which report the birth year as 2001.

[11] I exclude single parents for three reasons. First, given the more limited financial resources of single parents, they are likely to respond differently to changes in the generosity of benefits than their partnered counterparts. Second, since they have no partner to share the gender-neutral parental leave with, there is no consideration of allocation decisions, which is an important component of this analysis. Third, there is concern that their behavior may be influenced by other policy changes that occurred in that period, such as enhancements of the National Child Benefit that particularly targeted lower-income single parents. Small sample sizes preclude a separate analysis of single mothers as well.

[12] In supplementary regressions in Appendix A.3 I consider the impact of QPIP on conditional leave duration.

is unlikely to have been the case except for mothers who used it as a means to supplement the paid parental leave that they had exhausted. This is not a concern with the data on fathers' leave duration, since mothers are specifically asked about how many weeks of parental leave benefits their spouse has claimed or plans to claim.

4.2 Immediate Impact of QPIP

4.2.1 Regression Discontinuity Method

To evaluate the immediate impact of QPIP at the point that it was introduced, I adopt a sharp regression discontinuity (RD) design. Since the reform was introduced on 1st January 2006 with no gradual phase-in period, this provides a sharp cutoff after which a birth was eligible for QPIP. Moreover, there was limited certainty about the timing or the details of the reform until only a few months prior to its implementation.[13] The final details of QPIP, such as benefit amounts and the date of implementation, were only announced in mid-2005. Given that it takes some time to conceive a baby, it is reasonable to think that parents who gave birth around the cutoff were already pregnant at the time of announcement. Therefore, whether a birth occurred a few days prior to rather than a few days after January 1st 2006 was essentially random, allowing me to cleanly identify the local mean impact of QPIP through a regression discontinuity framework.

For each mother I have information on the year and month of birth of her youngest child, and the running variable for the RD is the distance in months from the cutoff date.[14] The model for each outcome is given by

$$Y_i = f(m_i) + \beta(m_i \geq Jan2006), \tag{1}$$

where $Y_{i,t}$ represents the outcome of mother i and m_i is the running variable which is the distance between the birth month and the cutoff of January 2006. β, the parameter of interest, represents the local mean impact of QPIP at the moment it was introduced. $m(x_i)$ is an unknown continuous function of the month of birth. I assume a flexible form for $f(m_i)$ and estimate it non-parametrically. I estimate equation (1)

[13] The idea of QPIP was discussed several years before the program came into place, but there were several bottlenecks in the policy process that prevented the program from being implemented. In June 2000, Quebec introduced legislation to establish its own parental leave program and in 2001 the Quebec National Assembly passed an Act that led to the development of a plan for Quebec's own program. However, the implementation of this legislation stalled because the federal government would not agree on the funds that the Quebec government would be able to keep in order to finance its own program. In an effort to force the federal government to act, the Quebec government asked the Quebec Court of Appeal to rule on the constitutionality of the EI provisions on maternity and parental benefits. Only once the court ruled that the Employment Insurance Act regarding maternity and parental benefits encroached on provincial jurisdiction and exceeded the powers of the Canadian Parliament, did negotiations begin between the two governments in 2004. It was not until the middle of 2005, more than four years after the initial act regarding the program had been passed, did the news emerge that the two governments had finally reached an agreement. QPIP officially came into place on the 1st of January 2006.

[14] The reported analyses use a bin width of one month. I also tried to widen the bins to 2-month intervals but rejected this method using an F-test since it over smoothed the data.

using local linear regressions (LLR) as Hahn et al. (2001) and Porter (2003) show that LLR performs better than kernel estimations at avoiding the boundary problem and obtaining a higher order of convergence at boundary points. The choice of bandwidth, i.e. the time window around the reform, is important since it determines the smoothing of the data and there is a tradeoff between variance and bias when choosing the optimal bandwidth. I select the bandwidth using the plug-in method proposed by Imbens and Kalyanaraman (2009). In Appendix A.1 I show my results are robust to the use of a parametric regression discontinuity analysis instead of a non-parametric approach. I provide the White (1980) heteroskedastic-consistent estimates of OLS standard errors, and in some specifications I allow for the clustering of standard errors within birthmonth, as suggested by Lee and Card (2008). .

To confirm internal validity I first verify that the pre-cutoff and post-cutoff group are balanced. Table II presents results from regression discontinuity analyses on the personal and household characteristics of the EICS Sample, showing no statistically significant discontinuities at the cutoff. The related graphs presented in Figures III and IV also confirm that observable household characteristics such as age of both parents, family size and number of children, as well as parents' educational characteristics, do not jump at the cutoff and should not confound any jump we detect in leave behavior. However, even though the discontinuities in sample characteristics are not statistically significant, a few are large enough to be economically significant, and therefore warrant concern. Accordingly, I present results from RD specifications with and without controls for these characteristics, to check that the program effect is not being confounded by changes in sample composition. In an RD-framework it is usually recommended that one ascertain that the density of births is similar on either side of the cutoff, to identify manipulation of the running variable. However, since the EICS data is collected as a stratified probability sample, it is not possible to run a McCrary (2008) density test.

4.2.2 Regression Discontinuity Results

Table III presents results from regression discontinuity analyses to identify the the immediate effect that QPIP had on parents' claim rates and duration. Panel I shows results for Quebec in a simple RD specification that does not control for any personal or household characteristics. Column 1 of Panel I reports that the introduction of QPIP in January 2006 is associated with a jump of 53.6 percentage points in the probability that a father claims parental leave benefits in Quebec. This point estimate is highly statistically and economically significant, representing more than 250% of the pre-reform participation rate of 21.3%. Column 2 indicates that QPIP resulted in a jump of 3.088 weeks in fathers' leave duration. This is also an economically and statistically significant effect, representing a 150% increase from the pre-reform average

of 2.011 weeks. For mothers' leave participation, the RD detects a jump of 15 percentage points, but the estimate is not statistically significant. It appears there is no immediate jump in mothers' leave duration at the cutoff. The inclusion of personal and household characteristics in the RD analysis do not affect the results, as can be seen in Panel II. This is reassuring, confirming that the program effect of QPIP detected by the RD is not being biased by sharp changes in sample composition. Panel III presents RD analyses where the standard errors have been clustered at the level of the monh of birth, and the only notable finding is that under this specification the jump in mothers' participation rates becomes statistically significant. For comparison purposes, Panel IV presents RD results for the control group of other provinces and show that there was no change in parents' behavior around the time of the reform in provinces that were not treated.

Figures 5-8 provide visual support for these results, graphing the local polynomial for each of the simple RD analyses. In Figures V and VI, For the treatment group we see clear discontinuities in both fathers' participation rates and leave duration at the cutoff, whereas no such jump exists in the corresponding graphs the control group. In Figure VII, we see that in the treatment group mothers' leave participation does jump at the cutoff but falls back down again. Figure VIII confirms there is no discontinuity in mothers' leave duration at the cutoff.

In aggregate, the RD results show that mothers' leave duration did not change while fathers' leave duration shot up. Furthermore, the increase in fathers' participation of over 50 percentage points, is considerably more than the share of families for whom QPIP loosened a binding constraint by extending total family leave. This evidence is therefore consistent with an intra-household flypaper effect, whereby the daddy quota induced participation from fathers even in families where the new 'daddy weeks' were essentially fungible.

4.2.3 Threats to Identification

The identification strategy in a sharp Regression Discontinuity framework depends crucially on the assignment to treatment being based on an exogenous measure. Since in this case assignment is based on the month of birth, clean identification requires that the timing of pregnancies and births is exogenous to the introduction of QPIP. Several studies have shown that decisions of fertility and timing of birth can respond to financial incentives (Gans and Leigh, 2009; Tamm, 2013). This leads to concerns that citizens may have known about QPIP sufficiently in advance and in detail in order to time their births so that they could utilize the new program. However, I present several pieces of supportive evidence that the strategic manipulation of births is not a significant concern confounding my estimates. First, details about the date and features of the reform were not officially announced until only a few months prior to its implementation. Figure IX presents

a 'Google Trends' graph tracking searches for the word 'QPIP' around the time of the reform.[15] There were relatively few searches for the program until January 2006 when QPIP came into place, consistent with the idea that details of QPIP were not commonly known sufficiently in advance of 2006 such that parents could plan their pregnancies accordingly.

Second, it is necessary to check whether residents of Quebec who were already pregnant when they learned of QPIP may have delayed their births until after January 2006 in order to be eligible for QPIP. Since RD analyses identify a jump at the cutoff and it is naturally infeasible to delay a birth by more than a few days, this is equivalent to checking that our RD estimates are not biased by pregnant women who were originally due in late December who may have been able to delay the delivery by a few days in order to qualify for QPIP instead of EI. As a check against this, I drop all observations in the one month surrounding the reform, and re-estimate the RD on this trimmed window to check how sensitive my results are to the exclusion of observations in the immediate vicinity of the cutoff (Barreca et al., 2011). Results from this 'trimmed' RD are shown in Appendix A.II and provide mostly consistent estimates, with the exception that the effect on fathers' leave duration appears smaller.

4.3 Average Treatment Effect of QPIP

4.3.1 Difference-in-differences Method

While the RD provides a clean estimate of the local mean impact of QPIP at the point at which it was introduced, it tells us nothing about whether QPIP continued to have an effect in the months and years to follow. To investigate the average treatment effect of QPIP over the period it has been in place, I use a longer span of data from 2002 to 2010, and employ a difference-in-differences method which exploits variation over provinces and time. It should be noted that another advantage of the difference-in-differences method is that there is less concern that it is biased by the manipulation of births immediately around the cutoff.

I estimate:

$$Y_i = \alpha + \beta Quebec_j * Post_t + \theta Post_t + \phi Z_{ijt} + \lambda_j + \delta_t + \epsilon_i, \tag{2}$$

where subscript i denotes the individual, subscript j denotes province and subscript t denotes the year. Y_i therefore represents the outcome of mother i. As outcomes, I explore whether the parent claims parental leave benefits and the duration of their actual or planned leave.[16] $Post_t$ is an indicator variable taking the

[15] Google Trends Searches using the full English and French names of the program present similar patterns.

[16] It should be noted that mothers were asked about spouse's leave duration only in survey years 2004 or later, leading to fewer observations for that outcome than those of fathers' leave participation. This does not affect my regression discontinuity framework because it only uses observations in a narrow window around the reform. Since I use the full 2002-2010 window for difference-in-differences, I conduct robustness checks where I use a uniform sample of observations from 2004-2010, and I find

value 1 if the year t is 2006 or greater, i.e., if the observation is from the post-reform period. The coefficient θ represents the change in the value of the outcome that is shared by all provinces. The term $Quebec_j * Post_t$ takes the value of 1 if the individual lives in Quebec in a post-reform year, and otherwise takes the value 0. The coefficient β therefore represents the DD estimate of primary interest as it captures the change in the value of the outcome post-reform that is unique to Quebec. Under the assumption that no other policy changes were enacted to affect it, β represents QPIP's average treatment effect on the treated (ATT).

The term Z_i is a vector of personal characteristics including age, education, legal marital status and immigrant status as well as household characteristics such as family size, number of children aged 0-1 and 1-5 and 6-17. Including these as regressors controls for changes in group composition. λ_j and δ_t denote the fixed province and year effects and ϵ_i is the error term..[17] I calculate cluster-robust standard errors that generalize the White (1980) heteroskedastic-consistent estimates of OLS standard errors to the clustered setting in order to account for possible heteroskedasticity and within-province dependence of standard errors, which are particularly a concern in difference-in-difference estimations since the regressor of interest is highly correlated within clusters (Bertrand et al., 2004). However, the small number of province-level clusters available in my sample leads to concerns regarding statistical inference since asymptotic tests have been shown to over-reject with too few clusters. Accordingly, I use the wild bootstrap-t procedures suggested by Cameron et al. (2008) to provide asymptotic refinement of standard errors. All analyses are conducted using ordinary least squares regressions despite the binary nature of some of the indicators because they resulted in very similar estimates as those from logit estimates.

It is important to discuss the assumptions under which this difference-in-differences identification strategy is valid, offering estimates of the true program effect. The first assumption is that no other programs or laws were enacted in Quebec at the same time which may have affected our outcome, such that the coefficient on $Quebec * Post - Reform$ may pick up effects of those other events instead. Notably, Quebec has a publicly subsidized childcare system while the rest of Canada does not. I verify that no changes were made to this program around the years of the QPIP reform. Further, I am careful to exclude data from before 2001, when the last expansion of the childcare program occurred. I also verify that Quebec did not make changes to child tax benefits or supplements around the time of the reform. The second necessary assumption of the difference-in-differences identification strategy is that of 'parallel trends' between the treatment and control group. That is, that the two groups should ideally have experienced similar trends prior to the introduction

consistent results (See Appendix A.6).

[17] I do not include controls for province-specific time trends as these would be highly collinear with the program effect of QPIP. As one would expect, in supplementary regressions I confirm that the inclusion of a Quebec-specific time trend absorbs some of the program effects, leading to smaller but consistent point estimates of QPIP's impact on leave behavior.

of the program, such that the control group offers a good proxy for the rate at which the outcome may have changed in the treatment province absent the treatment. I verify that this is the case for Quebec and other Canadian provinces in Figure X.

Table IV presents mean sample characteristics for the full 2002-2010 EICS sample as well as differences-in-means between the treatment and control groups over time. There are four difference-in-differences in characteristics which merit mention. First, the average age of new mothers grew more in Quebec than in other provinces, with a difference of 0.909 years. Second, the proportion of mothers that are legally married also grew more in Quebec than in other provinces. Third, the education levels of new parents changed more in Quebec than it did in other provinces, with an increase in mothers who have a high school education or less, and a decrease in mothers whose spouses only have a high school education or less. Lastly, though the difference is not statistically significant, the increase in proportion of immigrant mothers in Quebec by 6 percentage points may be economically significant. To account for these compositional changes, I present results from DD estimations with and without controlling for such personal characteristics as age, education and immigrant status of mother and spouse, as well as household characteristics such as family size and number of children aged 0-1, 1-5 and 6-17. Moreover, in Section 4.3.3 I discuss in detail whether these changes in sample composition could threaten identification, and I provide several robustness checks to allay any concerns.

4.3.2 Difference-in-differences Results

Table V presents results from difference-in-difference estimations of QPIP's impact on leave behavior. For fathers' leave outcomes, the DD estimates are very close to those obtained through the RD analysis. Column 1 of Panel I reports a significant program effect of 53.1 percentage points on fathers' participation rates and of 3.225 weeks in leave duration. For mothers' leave participation rates, the DD finds a program effect of 12.1 percentage points, or a 16% increase from the pre-reform baseline. Column 4 reports that QPIP is associated with an increase in mothers' leave duration of 2.84 weeks, though the estimate is not statistically significant. Panel II presents results from DD estimates that also control for personal covariates and province and year-fixed effects, and we see that this barely affects the point estimates, though it does affect inference in the case of fathers' leave duration. It is worth noting here that I also examined the program impact on leave duration conditional on participation but did not detect statistically significant effects (see Appendix Table A.III).[18]

[18] I find no significant negative effect on fathers' conditional leave duration. I find a small positive, though statistically insignificant, effect on mothers' conditional leave duration.

Note that, just as in the case of the RD, the DD results show that fathers' leave duration responded more strongly to QPIP than did mothers' leave duration. This is especially so when we consider the effects in relative terms: since mothers' took an average of 43 weeks of leave prior to the reform, a program effect of even 3 weeks would represent an increase of approximately 7%, whereas for fathers a program effect of 3 weeks represents an increase of 150%. This larger response in fathers' leave-taking to the reform is consistent with QPIP's daddy quota producing an intra-household flypaper effect due to a reduction in stigma cost.

To see this pattern in more detail, Table VI presents the impact of QPIP on the joint distribution of parental leave. Each cell represents a particular combination of mothers' and fathers' leave. The coefficients are estimated through difference-in-difference regressions where the outcome variable is an indicator for a family choosing that particular combination of mothers' and fathers' leave. Row A shows that QPIP reduced the likelihood of any combination where the father took 0 weeks of leave. Overall, it increased the likelihood that the father took between 1 and 5 weeks of leave, i.e. consumes from his quota. The coefficients in Column 5 are consistent with families responding to the relaxation of the total family leave constraint. The coefficients in Columns 3 and 4, however, find increases for fathers even in families that are not constrained by the cap, consistent with the idea that the daddy quota produced an intra-household flypaper effect.

Lastly, to explore whether QPIP affected income inequalities in leave-taking, Table VII reports the heterogeneous program effects of QPIP according to household income. I define a low-income household, alternately, to be one earning less than 20,000CAD, or one earning less than 40,000CAD in annual income.[19] Columns 1 and 2 of Table 7 suggest that QPIP produced a significantly smaller program effect on lower-income fathers than on fathers from higher-income households. For example, fathers from households earning less than 20,000CAD did increase their leave participation and duration under QPIP, but by 28.9 percentage points less and 4.2 weeks less than did fathers from higher-income households. This is an interesting finding because low-wage earners might be expected to respond more strongly to increased benefits: since benefits are capped, low-wage earners experience a larger marginal reduction in the price of leave. Moreover, the curious finding of a weaker response from low-income fathers is made doubly interesting by the fact that it is not mirrored by low-income mothers, who do respond more strongly than higher-income mothers, as expected. One possible explanation for why low-income fathers respond weakly to QPIP could be that they may have the highest stigma costs to taking leave, and that QPIP is less effective in reducing this stigma for them compared to higher-income fathers.[20] Overall, Table VII suggests that QPIP has increased income

[19] These thresholds correspond approximately to the 20th and 50th percentile of annual household incomes reported by mothers in the EICS (based on income in the month prior to birth). Changes in the coding of this variable over time make it impossible to use thresholds in between the two e.g. 30,000CAD.

[20] There exists considerable evidence of a positive association between education and more gender-egalitarian attitudes (Thornton and Freedman, 1979; Thornton et al., 1983; Shu, 2004) and that egalitarian gender attitudes trickle down through community

inequalities in leave-taking among fathers but has produced a reduction in the gap between low-income and higher-income mothers.

It should be noted that on net, even though fathers responded more strongly to QPIP than did mothers, the program effects did not eliminate gender differences in leave behavior. Even under QPIP, only 75% of fathers claimed parental leave benefits compared to 83% of mothers, and fathers on average took 5.46 weeks of leave, in stark comparison to mothers' average 46.37 weeks of leave. In aggregate QPIP was able to shrink but not complete close the gap in parental leave participation between men and women.

4.3.3 Threats to Identification

In difference-in-differences frameworks, it is common to challenge the exogeneity of reforms by questioning whether policies are in fact endogenously implemented as a response to trends in the outcome in the first place. For example, one may wonder whether policymakers in Quebec instituted a daddy quota because they were concerned by falling participation rates among fathers. Figure X plots fathers' participation rates in the treatment and control provinces over time. It shows that fathers' participation rates were not falling or rising more quickly in Quebec than other provinces prior to the reform. This confirms that the key assumption of the difference-in-differences identification, that of parallel trends between treatment and control groups, is satisfied. Prior to treatment both Quebec and other provinces experienced slightly increasing but parallel trends in fathers' participation. Thus, the control group offers a good proxy for the trajectory that Quebec would have followed absent the treatment, such that the program effects from DD regressions offer a good estimate of the level shift in participation rates due to the introduction of QPIP.

Another possibility I must consider is that even though details of QPIP were not available until mid-2005, the basic idea had been proposed in 2001 and citizens may have heard that a reform was being discussed that would offer generous incentives for fathers to participate. It is possible that couples who were particularly keen to have fathers take leave chose to delay pregnancy until the new program was in place, building pent-up demand that could only be released in early 2006. This seems unlikely as this would suggest families may have been willing to postpone a pregnancy by many years (five years passed between the program proposal and final announcement of implementation) in order to gain a few weeks of 'daddy' leave. Nevertheless, I check this by conducting an event study analysis to see how the program effect differed in the years following the reform. If my program effects were driven by pent-up demand which was released in 2006, we would expect fathers' uptake to jump up in 2006 but then to fall back down in later years once this pent-up demand was relieved. Figure XI presents results from an event-study analysis of the reform and shows that

education (Shu, 2004)

the program effect did not fall over time but remained constant or may have grown stronger as years passed.

Lastly, I consider the possibility that changes in the composition of the treatment group may be driving my program effects. I address these concerns in two ways. First, I conduct a summary test of composition bias by regressing leave outcomes on personal and household characteristics using the pre-treatment sample, obtaining predicted outcomes, and then running a simple difference-in-difference regression on the predicted outcomes (See Appendix Table A.4). The results for fathers' outcomes strongly support a causal interpretation of QPIP, since the reform has no 'program impact' on the predicted participation rates or durations. In the case of mothers' outcomes, there are small negative 'effects', suggesting that if anything, changes in sample composition should have biased us against finding a program effect of increased mothers' leave participation and duration. As a further robustness check, I conduct difference-in-difference analyses on subsamples of the data based on these characteristics, e.g., subsamples of non-immigrant mothers or young mothers etc (See Appendix Table A.5). The results for each subsample are very consistent to the results for the main sample, offering little reason to believe that QPIP's average program effects are largely driven by heterogeneous effects on these sub-groups that may have increased or decreased in prominence in the sample over time.

5 Long-Run Effects on Sex Specialization

5.1 Data on Parents' Time Use

To analyze the long-run effects of QPIP on the division of household labor, I use time-diary data from Canada's General Social Survey (GSS) (Statistics Canada, 2005, 2010). In this time-diary survey, the respondent is asked to record his or her activities as well as corresponding details such as locations and whether other people were present every 7 minutes over the 24-hour survey window. These data offer extremely precise measurements of time allocations, and are robust to response bias since the survey does not hint at gender issues. According to Kotsadaam and Finseraas (2012, pg 1619), who study the effect of a Norwegian daddy quota on the sharing of housework, "under ideal settings, we would exploit a time-use data-set with a large enough sample of individuals who had their last child in a time period around the reform to investigate the actual sharing, but no such data-set exists" (for Norway). Fortunately, exactly such a data-set exists for Canada, and I use it in this study.

I specifically use the two most recent rounds of the GSS that collected time-diary data: cycle 19 that was conducted in 2005 and cycle 24 that was conducted in 2010. Since QPIP was introduced in Quebec in 2006, observations from 2005 are considered to be in the pre-reform period while observations from 2010

are considered to be in the post-reform period. The target population of the GSS includes all persons 15 years of age and older in Canada excluding full-time residents of institutions. Approximately one-fifth of the observations are from Quebec, the treatment group, while the rest of the observations come from the control group that consists of other provinces in Canada.[21] The sample comprises parents aged 18-50 whose youngest child is aged between 1-8 and who are in a married or cohabitating relationship. Parents whose youngest child is under one year old are excluded from the sample to eliminate the possibility they may still be on parental leave at the time of the survey. In all analyses I exclude parents whose youngest child is aged 4. Since the survey does not collect information on the child's exact date of birth, it is impossible to determine whether parents who were interviewed in Quebec in 2010 experienced this birth in 2005 or 2006, and therefore whether or not they were exposed to the QPIP.

I measure time spent in various types of work in minutes per day as recorded by their time-diary. For parents' market outcomes, I investigate the time spent in paid work (which includes commuting to paid work) and time spent physically at the workplace, as well as labor market outcomes such as annual income, employment status, full-time employment, usual weekly hours, and weeks worked last year. In supplementary analyses I also look at hourly wages, but in these regressions the sample sizes drop significiantly and so I limit my interpretation of these results as only suggestive. It should be noted that some of the measures for labor market outcomes, e.g. full-time employment and hourly wages, are constructed carefully using information from other variables (For details of the construction methods please see Appendix B).

For parents' non-market outcomes, I examine total time spent in non-market work (the sum of housework and childcare) as well as childcare and housework separately, and total time spent physically at home and time spent in the vicinity of family members. Following Aguiar and Hurst (2007), I define total domestic work as the sum of time in 'core' non-market work (such as meal preparation and cleanup, laundry, ironing, dusting, and indoor cleaning), time spent obtaining goods and services (such as shopping for groceries or household supplies) and 'other' home production such as maintenance and repairs, gardening, and caring for houseplants and pets. Following Craig (2006), I define total childcare as the sum of time spent in routine childcare (such as feeding children or getting them ready for school), interactive childcare (such as helping with homework or reading to children) and also travel and communication related to childcare such as driving children to school or attending a parent-teacher conference. I also look at "total time spent at home," i.e. the sum of all time that the respondent identifies his location as his residence, and "total time spent with family members," which is the sum of all time where the respondent notes a family member was also present.

[21] Specifically, the control group comprises observations of parents resident in Newfoundland and Labrador, Prince Edward Island, Nova Scotia, New Brunswick, Ontario, Manitoba, Saskatchewan, Alberta, British Columbia, Yukon, Nunavut and the Northwest Territories.

It is important to note that these two outcomes could include time spent sleeping, which could be recorded "at home" or "in the presence of a family member" without necessarily indicating any contribution to home production or 'quality' time with family. Lastly, I explore non-market production at a greater level of detail, e.g., breaking childcare up into interactive and routine activities separately, and breaking domestic work into several categories such as maintenance/repairs, cooking, shopping, housekeeping, and other chores.

There are two limitations of the GSS data that merit mention. First, since the data are not collected at the couple-level, I cannot track changes in spouses' behavior within the same household; instead, my results show how mothers' and fathers' behavior changed on average across households. A second limitation is that given that families can move between provinces, it is possible that families observed in Quebec experienced their last birth in one of the other provinces or vice versa. However, given that the 2006 census reported less than 3% of the Canadian population moving between provinces, the number of recent-movers in and out of Quebec in my sample is expected to be small and unlikely to bias the results significantly. Moreover, since this type of cross-contamination should reduce differences between observations in Quebec and other provinces, my results would underestimate the true causal effect of paternity leave on long-term behavior.

5.2 Double- and Triple-Differencing Methods

To identify the long-term causal effects of QPIP, I apply the same difference-in-differences method described in equation (2) of section 4.3.1 to the time-use data. I use a GSS sample of parents whose youngest child is aged 1-3. Just as before, I focus on the coefficient on "Quebec*Post" as the DD effect of exposure to the QPIP program.

However, since Time Use GSS data is only available every 5 years, the above method compares changes between 2005 and 2010 among Quebecois parents of children aged 1-3, compared to identical parents in other provinces. This gives rise to the concern that something else may have changed over that period in Quebec such that a simple double-difference could confound a Quebec-wide trend with a change in behavior causally related to the QPIP program. To mitigate such concerns, I devise a robustness check that utilizes a placebo group of parents whose youngest child is aged 5-8.[22] These parents form a convenient placebo group because even if they are observed in the treated province in the post-treatment year (2010), their children are slightly too old for them to have been eligible for QPIP. I use a difference-in-difference-in-differences (DDD) identification strategy that exploits variation in exposure to paternity leave across provinces, time, and age-group of the child. In this triple-differencing setup, a parent is only considered to be exposed to

[22] I vary the age restrictions on the placebo group, using the same minimum age of 5 but increasing the maximum from 8 up till 14 and get similar results.

QPIP if they are observed in Quebec in 2010 and their youngest child was born since 2006, i.e., the child is aged 1-3.

For each triple-differencing regression I estimate the following equation:

$$\begin{aligned} Y_i = \alpha + \delta Quebec_j * Post_t * Under3_g + \beta Quebec_j * Post_t + \sigma Under3_g * Post_t \\ + \theta Quebec_j * Under3_g + \gamma Post_t + \chi Under3_g + \phi Z_{ijt} + \lambda_j + \epsilon_i, \end{aligned} \tag{3}$$

where subscript i denotes the individual, subscript j denotes province and subscript t denotes the year. Y_{ijt} represents the outcome for parent i in province j in year t. $Post_t$ is an indicator variable taking the value 1 if the observation is from after 2006, i.e., if the observation occurred after the reform was introduced in Quebec. The coefficient γ represents the change in the value of the outcome that is shared by all provinces. An interaction term, $Quebec_j * Post_t$, is included to capture the change in the value of the outcome post-reform that is unique to Quebec. $Under3_g$ is an indicator taking value 1 when the parents' youngest child is aged 1-3 years old, and 0 if the child is aged 5-8 years old. The parameter of interest is δ, the coefficient on $Quebec_j * Post_t * Under3_g$, which captures the effect of being in the treated province in the post-treatment period and having a child young enough that the parent was eligible for the treatment. The term Z_{ijt} is a vector of personal characteristics including age, spouse's age, marital status, nation of birth, as well as household characteristics such as family size, number of children aged 0-14, and age of youngest child.[23] I control for Province fixed effects through the term λ_j. ϵ_{ijt} is an i.i.d error term. Here again I calculate heteroskedasticity-robust standard errors which are clustered at the province level.

The inclusion of the placebo group of parents of slightly older children offers an ideal robustness check as it differences out changes in Quebec over time that are unrelated to exposure to QPIP. For example, regional economies may have fared differently in the recent recession, but the parents of children of different ages within the same province would have faced the same economic opportunities. Moreover, if one were concerned that the introduction of QPIP is endogenous to a Quebecois culture that increasingly values gender equality, then including Quebecois fathers of slightly older children will control for trends in egalitarian beliefs in Quebec. Therefore, when evaluating the validity of my main DD results one can use the following protocol. The validity of DD results as a causal link is supported whenever the triple-difference results have the same sign and a similar or larger magnitude. But in cases where the Triple difference results have an opposite sign or a smaller magnitude, we must be careful in interpreting the DD results as they may just reflect picking up

[23] I do not control for educational characteristics for two reasons. First, for the education questions the GSS Time Use survey has lower response rates, imposing significant restrictions on sample size. Second, leaving education controls out of the regressions allows the education of each parent to be endogenous.

Quebec-specific trends. In my discussion of the results in Section 6, I am careful to only highlight changes in parents' behavior which are supported by both the double-difference and triple-difference results.

Table VIII presents sample characteristics for the GSS data across treatment, control and placebo groups in the years 2005 and 2010, as well as the differences across the groups over time. Reassuringly, I detect only one significant difference across the groups over time: between 2005 and 2010 the proportion of fathers who were not born in Canada grew more rapidly in the 'exposed' group (i.e. fathers in Quebec with a youngest child aged 1-3) than in the other groups. However in every other characteristic such as age, spouse's age, number and age of children, family size, there are no reasons to believe the triple-differencing identification strategy would be mistakenly picking up changes in group composition. Nevertheless, I include controls for these parental and household characteristics in each triple-differencing regression.

5.3 Double- and Triple-Difference Results

Before beginning my discussion of the long term effects of QPIP, it is important to first establish a baseline for how households behaved prior to the existence of QPIP. Table IX shows mean outcomes for parents with youngest children aged 1-3 in the year 2005; it shows that household responsibilities were clearly divided along gendered lines. Mothers spent more time in non-market work including housework and childcare, especially in time-inflexible chores such as cooking, housekeeping and routine childcare. The only household chore in which fathers spend more time than mothers is that of maintenance and repairs, which is flexible, not routine and in line with norms of masculinity. The ratio is reversed when we consider market work and labor market outcomes. Fathers spend considerably more time in paid work and physically at the workplace. Moreover, fathers are more likely to be employed, full-time workers, work longer hours per week, work more weeks per year, and earn more than mothers both in terms of annual personal income. A clear pattern of sex specialization within the household is evident, providing us with a baseline against which to evaluate the magnitude of any program effects.

Table X presents results exploring parents' involvement in market production. It shows results from both double-difference regressions using the sample of parents whose child is aged 1-3, and to verify robustness it also shows results from triple-difference regressions using the full sample which includes 'placebo parents' whose child is aged 5-8. For the sake of brevity, I will focus my discussion on only those results from the double-difference regressions where the sign and significance of the coefficients are supported by the results from triple-difference regressions. Panel I shows the results for fathers investments in market work. The DD cannot detect statistically significant changes in fathers' market outcomes, though some of the coefficients do show decreases that are large enough to be economically significant, and are supported by the triple-

difference results. For example, there is suggestive evidence that exposed fathers spend considerably less time in paid work per day, and experience a not insignificant reduction in personal income. Mothers, on the other hand, experience impressive gains from exposure to QPIP. Exposed mothers report more time in paid work and more time physically at the workplace per day. Further, the DD detects a rise in exposed mothers' annual income of 5,819CAD - an increase of over 25% of baseline income. Conditional on being employed, the DD analysis finds exposed mothers are 4.6 percentage points more likely to be full-time employed, and report working 1.35 more hours per week. In sum, Table X presents compelling evidence of reduced male specialization in market work.[24]

In Table XI, I present results for parents' time in non-market work and related outcomes. The DD finds that exposed fathers spend 37 minutes longer in non-market work per day, representing a 23% increase from baseline. In housework, the DD finds exposed fathers spend 15 minutes longer per day than their counterparts, a 21% increase from baseline. In addition, exposure to QPIP is associated with an increase of 36 minutes spent physically at home, and 29 minutes more in the immediate vicinity of family members. Interestingly, exposure to QPIP is also associated with increased time spent by mothers in non-market work, although the absolute and relative magnitude of their increase is smaller than that of fathers. The DD results show that exposed mothers reduce their time in housework by 18 minutes and increased their time in childcare by 48 minutes, leading to an increase of 30 minutes in total non-market work (10% increase from baseline). Exposure is associated with mothers spending 30 fewer minutes spent physically at home.

Since exposed fathers increase non-market work more than do exposed mothers, Table XI does suggest that in aggregate female specialization in home production is reduced. Nevertheless, it is interesting that I detect an increase at all in mothers' childcare rather than a decrease. There are several possible explanations for this, though the data do not allow me to detangle these mechanisms. First, as previous research has suggested, mothers may be less willing to reduce time in childcare than in other household duties (Craig, 2005), so when paternity leave induces fathers to increase non-market contributions it may be efficient for them to increase time in housework that is less preferred by mothers. Alternatively, both parents may have equal preference for childcare but exposed mothers have gained bargaining power that they use to negotiate away from less-preferred housework and towards more-preferred childcare. Lastly, a possible explanation is that although parents' relative productivities in market versus non-market work may have changed, within the realm of non-market work mothers still have a comparative advantage in childcare versus housework.

Since I exploit variation in exposure to QPIP rather than actual participation, I provide evidence on

[24] Supplementary analyses which explore wages suggest the changes in parents' earnings are due to not only reduced time in work but also reduced returns to work - the results in Appendix Table A.VII suggest QPIP is associated with a reduction in fathers' hourly wages of 16% and an increase in mothers' hourly wages of 49%.

'intent-to-treat' (ITT) estimates that are preferable to estimates of 'treatment on the treated' (TOT) for several reasons. First, TOT estimates could be subject to the same bias from selection into treatment that previous cross-sectional studies have been criticized for. Second, from a policy-making perspective, ITT effects may be more relevant as they allow for feedback effects whereby the 'daddy quota' could have changed expectations and norms over and above the effects of actually using the leave option. The 'daddy quota' sent a strong public message about the importance of fathers' involvement in the home, which may have incentivized fathers who were exposed to QPIP but not treated to nevertheless change their behavior. Furthermore, a change in the behavior of treated households may change costs and incentives for neighboring households. For example, workplace expectations for all mothers may rise as treated mothers increase their career commitment and consequently, the penalty may increase for the untreated mothers who do not. Nevertheless, it is safe to assume that feedback effects on parents who were exposed but not treated are smaller than the first-order effects on parents who were treated, such that the ITT results presented here provide a lower bound for the causal effects of paternity leave on those who actually take it.

5.4 Threats to Identification

An important issue that must be addressed is that of changes in fertility. Since QPIP provided greater financial incentives to have children, it may have led to an increase in fertility rates in Quebec. This gives rise to the concern that the long-term program effects I detect may be driven by the marginal couples that are induced to have a child by the new program. I address this concern in multiple ways. First, it should be noted that the marginal couples do not appear different in any observable way, since I do not find significant changes in sample characteristics in Quebec compared to other provinces (see Table VIII). For example, one could hypothesize that due to the generosity of QPIP's benefits people were concerned that the program was temporary and so couples may have rushed to have a baby sooner than they would otherwise. If this were the case, one would expect to see a unique decrease in the age of parents in Quebec, but Table VIII shows this was not so. Moreover, I control for personal and household characteristics in all my regressions, so the program effects I detect should not be biased by Quebecois households having larger families or more young children in the post-reform period. Second, the marginal couples may be different in some unobservable characteristic, e.g. egalitarianism, which I cannot control for. However, it should be noted that the summary statistics of observable characteristics of births described in Table IV suggests precisely the opposite: if anything, mothers who experience a birth in post-QPIP Quebec are more likely to be older, legally married, less educated and immigrants - all characteristics that are associated with less gender-egalitarianism. Third, if we consider that QPIP's effects on leave behavior were heterogeneous

based on beliefs, it becomes unclear what direction this bias, if it exists, would take. On the one hand, one could imagine that the fathers most responsive to paternity leave incentives are those who are more open-minded, which would bias my study in favor of finding reduced sex specialization. However, since we saw that most families did not exhaust their leave prior to QPIP, these open-minded fathers were always free to take leave. On the other hand, one could argue that QPIP's reformed features targeted more traditional families, which would bias my study against finding reduced sex specialization. For example, the increase in income replacement provided the greatest marginal benefit to low-income parents, who are more likely to have traditional beliefs. Also, QPIP's daddy-quota provided stronger incentives for fathers who were previously financially dis-incentivized from participating - e.g., those who earned significantly more than their wives - who are also more likely to have traditional beliefs. Therefore, it cannot be known which direction this fertility bias, if it exists, would take, and an argument can be made that it may lead me to under-estimate the true program effect of QPIP on sex specialization.

6 Conclusion

This paper provides the first comprehensive study of the short- and long-run effects of the QPIP reform and offers an important contribution to the literature on paternity leave. It is the first to provide causal evidence that daddy quotas may influence behavior even when they do not relax a binding constraint, suggesting that the 'daddy-only' label produces an intra-household flypaper effect that makes leave stick to fathers. This study also offers the first comprehensive causal analysis of the effect of paternity leave on the household division of labor. It provides strong evidence that by altering the initial distribution of parenting responsibilities, paternity leave can influence decisions about how to allocate parents' resources to childcare, domestic work and paid work in later years.

The multi-dimensional nature of the effect that I document here has several implications for research on this topic. First, the program effects detected in different outcomes highlight the importance of exploring the full range of both parents' time commitments simultaneously. For example, studies that report that paternity leave does not improve sharing of childcare, such as that of Kluve and Tamm (2009) and Ekberg et al. (2013), should be considered alongside my finding that it does improve sharing of housework. Also, my finding that both parents' increase non-market contributions under QPIP highlights the importance of using data and measures which could capture this, as opposed to measures such as relative proportions that would not be able to detect that mothers' and fathers' time in an activity may change simultaneously.

The results of this study also have important policy implications. First, they suggest that 'daddy-only'

quotas may help fathers overcome such barriers to taking leave as social stigma and perceived professional penalties. Second, they suggest that it is possible for policies that induce changes in short-term behavior to have persistent effects on people's behavior in the long term, i.e., that a reform resulting in an increase in fathers' leave duration of 3 weeks could be sufficient to stimulate a shift in household dynamics for years to come. Third, and perhaps most importantly, my results suggest that there need not be a trade-off between gender equality and parental investments in children, such that paternity leave may present us with a rare win-win scenario.

References

Aguiar, M. and Hurst, E. 2007. Measuring trends in leisure: The allocation of time over five decades. *The Quarterly Journal of Economics*, 122(3):969–1006.

Akerlof, G. A. and Kranton, R. E. 2010. *Identity Economics: How Our Identities Shape Our Work, Wages, and Well-Being*. Princeton University Press, Princeton, NJ.

Allen, S. and Daly, K. 2007. The effects of father involvement: An updated research summary of the evidence inventory. Report., University of Guelph: Centre for Families, Work Well-Being.

Barreca, A. I., Guldi, M., Lindo, J. M., and Waddell, G. R. 2011. Saving babies? revisiting the effect of very low birth weight classification. *The Quarterly Journal of Economics*, 126(4):2117–2123.

Becker, G. 1981. *A treatise on the family*. Harvard University Press, Cambridge, Mass.

Bertrand, M., Duflo, E., and Mullainathan, S. 2004. How much should we trust differences-in-differences estimates? *The Quarterly Journal of Economics*, 119(1):249–275.

Bianchi, S. M. 2011. Family change and time allocation in american families. *Annals of the American Academy of Political and Social Science*, 638(1):21–44.

Bianchi, S. M. 2012. Housework: Who did, does or will do it, and how much does it matter? *Social Forces*, 91(1):55–63.

Blair, S. L. and Lichter, D. T. 1991. Measuring the division of household labor: Gender segregation of housework among american couples. *Journal of Family Issues*, 12(1):91–113.

Blau, F. D. and Kahn, L. M. 2007. Changes in the Labor Supply Behavior of Married Women: 19802000. *Journal of Labor Economics*, 25:393–438.

Blau, F. D. and Kahn, L. M. 2013. Female labor supply: Why is the united states falling behind? *American Economic Review*, 103(3):251–56.

Blau, F. D., Ferber, M. A., and Winkler, A. E. 2014. *The economics of women, men, and work*, volume 4. Pearson, Boston, MA, 7th edition edition.

Boll, C., Leppin, J., and Reich, N. 2014. Paternal childcare and parental leave policies: evidence from industrialized countries. *Review of Economics of the Household*, 12(1).

Brandth, B. and Kvande, E. 1998. Masculinity and child care: The reconstruction of fathering. *The Sociological review*, 46(2):293.

Bruning, G. and Plantenga, J. 1999. Parental leave and equal opportunities: experiences in eight european countries. *Human Resources Abstracts*, 34(4).

Bygren, M. and Duvander, A.-Z. 2006. Parents workplace situation and fathers parental leave use. *Journal of Marriage and Family*, 68(2):363–372.

Cameron, A. C., Gelbach, J. B., and Miller, D. L. August 2008. Bootstrap-Based Improvements for Inference with Clustered Errors. *The Review of Economics and Statistics*, 90(3):414–427.

Card, D. and DiNardo, J. E. February 2002. Skill biased technological change and rising wage inequality: Some problems and puzzles. Working Paper 8769, National Bureau of Economic Research.

Cools, S., Fiva, J. H., and Kirkebøen, L. J. 2011. Causal Effects of Paternity Leave on Children and Parents. CESifo Working Paper Series 3513, CESifo Group Munich.

Craig, L. 2006. Does father care mean fathers share? a comparison of how mothers and fathers in intact families spend time with children. *Gender and Society*, 20(2):pp. 259–281.

Craig, L. January 2005. How do they do it? SPRC Discussion Papers 136, Social Policy Research Center.

Dahl, G. B., L?ken, K. V., and Mogstad, M. 2014. Peer Effects in Program Participation. *American Economic Review*, 104(7):2049–74.

del Carmen Huerta, M., Adema, W., Baxter, J., Han, W.-J., Lausten, M., Lee, R., and Waldfogel, J. January 2013. Fathers' Leave, Fathers' Involvement and Child Development: Are They Related? Evidence from Four OECD Countries. OECD Social, Employment and Migration Working Papers 140, OECD Publishing.

Doucet, A., Tremblay, D. G., and Lero, D. *Leave Policy and Research: Canada*. International Review of Leave Policies and Related Research. Employment Relations Research, London, 5th edition edition, 2010.

Duvander, A.-Z. and Johansson, M. 2012. What are the effects of reforms promoting fathers parental leave use? *Journal of European Social Policy*, 22(3):319–330.

Duvander, A.-Z., Lappegrd, T., and Andersson, G. 2010. Family policy and fertility: fathers' and mothers' use of parental leave and continued childbearing in norway and sweden. *Journal of European Social Policy*, 20(1):45–57.

Ekberg, J., Eriksson, R., and Friebel, G. 2013. Parental leave a policy evaluation of the swedish daddy-month reform. *Journal of Public Economics*, 97(0):131 – 143.

El Nokali, N. E., Bachman, H. J., and Votruba-Drzal, E. 2010. Parent involvement and childrens academic and social development in elementary school. *Child Development*, 81(3):988–1005.

Estes, S. B. 2011. How are workfamily policies related to the gendered division of domestic labor? *Sociology Compass*, 5(3):233–243.

Evertsson, M. and Duvander, A.-Z. 2011. Parental leavepossibility or trap? does family leave length effect swedish womens labour market opportunities? *European Sociological Review*, 27(4):435–450.

Fuwa, M. and Cohen, P. N. 2007. Housework and social policy. *Social science research*, 36(2):512–530.

Gans, J. S. and Leigh, A. 2009. Born on the first of july: An (un)natural experiment in birth timing. *Journal of Public Economics*, 93(12):246 – 263.

Haas, L. and Hwang, P. *Parental leave in Sweden. Volume 35.* Parental leave: Progress or pitfall? Research and policy issues in Europe. NIDI/CGBS Publications, The Hague Brussels, 1999.

Haas, L. 1990. Gender equality and social policy: Implications of a study of parental leave in sweden. *Journal of Family Issues*, 11(4):401–423.

Haas, L. and Hwang, C. P. 2008. The impact of taking parental leave on fathers participation in childcare and relationships with children: Lessons from sweden. *Community, Work amp; Family*, 11(1):85–104.

Haas, L. and Rostgaard, T. 2011. Fathers' rights to paid parental leave in the nordic countries: consequences for the gendered division of leave. *Community, Work amp; Family*, 14(2):177–195.

Hahn, J., Todd, P., and Van der Klaauw, W. January 2001. Identification and Estimation of Treatment Effects with a Regression-Discontinuity Design. *Econometrica*, 69(1):201–09.

Han, W.-J., Ruhm, C., and Waldfogel, J. December 2007. Parental leave policies and parents' employment and leave-taking. Working Paper 13697, National Bureau of Economic Research.

Hersch, J. and Stratton, L. S. 2002. Housework and wages. *The Journal of Human Resources*, 37(1):pp. 217–229.

Hochschild, A. R. and Machung, A. 1989. *The second shift : working parents and the revolution at home*. Viking, New York, N.Y.

Hook, J. L. 2006. Care in context: Men's unpaid work in 20 countries, 1965-1998. *American sociological review.*, 71(4):639.

Hook, J. L. 2010. Gender inequality in the welfare state: sex segregation in housework, 1965-2003. *AJS; American Journal of Sociology*, 115(5):1480–523.

Imbens, G. and Kalyanaraman, K. February 2009. Optimal bandwidth choice for the regression discontinuity estimator. Working Paper 14726, National Bureau of Economic Research.

Johansson, E.-A. March 2010. The effect of own and spousal parental leave on earnings. Working Paper Series 2010:4, IFAU - Institute for Evaluation of Labour Market and Education Policy.

Juhn, C. and Murphy, K. M. 1997. Wage Inequality and Family Labor Supply. *Journal of Labor Economics*, 15(1):72–97.

Kluve, J. and Tamm, M. 2009. Now daddy's changing diapers and mommy's making her career: Evaluating a generous parental leave regulation using a natural experiment. IZA Discussion Papers 4500, Institute for the Study of Labor (IZA).

Kotsadam, A. and Finseraas, H. 2012. The state intervenes in the battle of the sexes: Causal effects of paternity leave. *Social science research*, 40(6):1611–1622.

Kotsadam, A., Finseraas, H., and Ugreninov, E. *The long-term effect of Parental Leave on Mothers' Earnings*. PhD thesis, University of Gothenberg, 2011.

Lalive, R. and Zweimller, J. 2009. How does parental leave affect fertility and return to work? evidence from two natural experiments. *The Quarterly Journal of Economics*, 124(3):1363–1402.

Lamb, M. E. 2010. *The Role of Father in Child Development*. Wiley Press, Hoboken, NJ.

Lee, D. S. and Card, D. 2008. Regression discontinuity inference with specification error. *Journal of Econometrics*, 142(2):655 – 674.

Lee, D. S. and Lemieux, T. 2010. Regression discontinuity designs in economics. *Journal of Economic Literature*, 48(2):281–355.

Lemieux, T. 2002. Decomposing changes in wage distributions: A unified approach. *The Canadian Journal of Economics*, 35(4):pp. 646–688.

Lundberg, S. and Pollak, R. A. 1996. Bargaining and distribution in marriage. *The Journal of Economic Perspectives*, 10(4):pp. 139–158.

McCrary, J. February 2008. Manipulation of the running variable in the regression discontinuity design: A density test. *Journal of Econometrics*, 142(2):698–714.

Moss, P. and O'Brien, M. 2006. International review of leave policies and related research 2006. *Employment Relations Research Series*, (52).

Nepomnyaschy, L. and Waldfogel, J. 2007. Paternity leave and fathers' involvement with their young children. *Community, Work ; Family*, 10(4):427–453.

O'Brien, M. 2009. Fathers, parental leave policies, and infant quality of life: International perspectives and policy impact. *The Annals of the American Academy of Political and Social Science*, 624(1):190–213.

OECD. December 2012. Closing the gender pay gap: Act now. Technical Report ISBN 978-92-64-17936-3, Organization for Economic Cooperation and Development.

Porter, J. 2003. Estimation in the regression discontinuity model.

Pylkkänen, E. and Smith, N. March 2003. Career interruptions due to parental leave: A comparative study of denmark and sweden. Technical report, OECD Publishing.

Rege, M. and Solli, I. 2013. The impact of paternity leave on fathers future earnings. *Demography*, 50(6): 2255–2277.

Rieck, K. M. E. June 2012. Does Child Care Affect Parents Sickness Absence? Evidence From A Norwegian Paternity Leave Reform. Working papers in economics, University of Bergen, Department of Economics.

Sanchez, L. and Thomson, E. 1997. Becoming mothers and fathers: Parenthood, gender, and the division of labor. *Gender and Society*, 11(6):pp. 747–772.

Schönberg, U. and Ludsteck, J. March 2007. Maternity Leave Legislation, Female Labor Supply, and the Family Wage Gap. IZA Discussion Papers 2699, Institute for the Study of Labor (IZA).

Sevilla, A. 2014. On the importance of time diary data and introduction to a special issue on time use research. *Review of Economics of the Household*, 12(1):1–6.

Shu, X. 2004. Education and gender egalitarianism: The case of china. *Sociology of Education*, 77(4):pp. 311–336.

Statistics Canada. 2002-2010. Employment insurance coverage survey. [Restricted-access Master Files], Ontario: Statistics Canada [Producer and Distributor].

Statistics Canada. 2005, 2010. General social survey. [Restricted-access Master Files], Ontario: Statistics Canada [Producer and Distributor].

Sullivan, O., Coltrane, S., Mcannally, L., and Altintas, E. 2009. Father-friendly policies and time-use data in a cross-national context: Potential and prospects for future research. *The ANNALS of the American Academy of Political and Social Science*, 624(1):234–254.

Tamis-LeMonda, C. and Cabrera, N. 2002. *Handbook of Father Involvement: Multidisciplinary Perspectives*. Erlbaum Press, Mahwah, NJ.

Tamm, M. 2013. The impact of a large parental leave benefit reform on the timing of birth around the day of implementation*. *Oxford Bulletin of Economics and Statistics*, 75(4):585–601.

Tanaka, S. and Waldfogel, J. 2007. Effects of parental leave and work hours on fathers involvement with their babies. *Community, Work Family*, 10(4):409–426.

Thornton, A. and Freedman, D. 1979. Changes in the sex role attitudes of women, 1962-1977: Evidence from a panel study. *American Sociological Review*, 44(5):pp. 831–842.

Thornton, A., Alwin, D. F., and Camburn, D. 1983. Causes and consequences of sex-role attitudes and attitude change. *American Sociological Review*, 48(2):pp. 211–227.

Ugreninov, E. 2013. Can family policy reduce mothers sick leave absence? a causal analysis of the norwegian paternity leave reform. *Journal of Family and Economic Issues*, 34(4):435–446.

White, H. 1980. A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4):817–38.

Zhelyazkova, N. 2013. Fathers' use of parental leave. What do we know? MERIT Working Papers 022, United Nations University - Maastricht Economic and Social Research Institute on Innovation and Technology (MERIT).

Table I: Details of Parental Leave Programs in Canada

	Employment Insurance	QPIP Basic Plan	QPIP Special Plan
Eligibility	600 hours of insurable employment	2000CAD of insurable earnings	2000CAD of insurable earnings
Basic Replacement Rate	55%	70% for all maternity, & paternity leave, first seven weeks of parental leave and 55% thereafter	75%
Max insurable earnings	39,000CAD	57,000CAD	57,000CAD
Waiting Period	2 weeks	None	None
Duration	Total 50 weeks = 15 weeks maternity leave + 35 weeks parental leave + no paternity leave	Total 55 weeks = 18 weeks maternity leave + 32 weeks parental leave + 5 weeks paternity leave	Total 40 weeks = 15 weeks maternity leave + 25 weeks parental leave + 3 weeks paternity leave

Source: Table constructed by author using information from the Digest of Benefit Entitlement Principles, available at http://www.servicecanada.gc.ca/eng/ei/digest/chp12_appendix.shtml. For features which may change on a yearly basis, such as the amount of maximum insurable earnings, figures provided are for 2006.

Table II: Regression Discontinuities in Personal and Household Characteristics in EICS data

	Mother's Age	Fathers' Age	Children aged 0-1	Children aged 1-5	Family Size	Immigrant
RD Estimate	0.138	0.624	0.012	-0.130	-0.324	0.110
	[0.894]	[0.610]	[0.340]	[0.432]	[0.118]	[0.410]
Bandwidth (months)	16.69	17.14	4.06	9.65	9.80	8.147

	Mother's Educ $\leq$ High School	Mother's Educ = Some College	Mother's Educ = College	Fathers' Educ $\leq$ High School	Fathers' Educ = Some College	Fathers' Educ = College
RD Estimate	0.014	-0.100	0.091	-0.101	0.222	-0.123
	[0.913]	[0.540]	[0.484]	[0.264]	[0.191]	[0.464]
Bandwidth (months)	8.46	8.94	8.80	8.28	8.94	8.76

Notes: Table presents results from non-parametric RD analysis of Quebec, using local linear regressions to detect discontinuities in personal and household characteristics between parents who experienced a birth before versus after the cutoff of January 2006. Heteroskedasticity-robust p-values are presented in square brackets. The optimal bandwidth was selected using the plug-in procedure suggested by Imbens and Kalyanaraman (2009). Sample spans 2004-2007 of the EICS data and comprises mothers aged 18-40 in cohabitating or married relationships who have experienced a birth in the last year.

Table III: Local Mean Impact of QPIP on Parents' Leave Behavior

OUTCOMES:	(1) Fathers' Participation Rates	(2) Fathers' Leave Duration (Weeks)	(3) Mothers' Participation Rates	(4) Mothers' Leave Duration (Weeks)
Baseline for Quebec (2002-2005 Average)	0.213	2.011	0.725	43.140
I. RD analysis of Quebec				
Jump at Cutoff	0.536***	3.088*	0.155	-0.277
	[0.00]	[0.06]	[0.18]	[0.93]
Bandwidth (months)	8.692	18.968	8.568	24.874
II. RD analysis of Quebec, including personal covariates				
Jump at Cutoff	0.531***	3.126**	0.188	0.930
	[0.00]	[0.04]	[0.15]	[0.93]
Bandwidth (months)	8.791	18.968	8.568	24.874
III. RD analysis of Quebec, including personal covariates and errors clustered by month of birth				
Jump at Cutoff	0.530***	3.125**	0.187**	0.930
	[0.00]	[0.01]	[0.01]	[0.78]
Bandwidth (months)	8.692	18.968	8.568	24.874
IV. RD analysis of control provinces				
Jump at cutoff	-0.006	-1.332	-0.008	-0.207
	[0.92]	[0.11]	[0.93]	[0.93]
Bandwidth (months)	6.143	14.733	7.11	20.669

*** p<0.01, ** p<0.05, * p<0.1, Heteroskedasticity-robust p-values in parentheses

Notes: Table presents results from non-parametric RD analysis of Quebec, using local linear regressions to detect discontinuities in leave outcomes between parents who experienced a birth before versus after the cutoff of January 2006. The running variable is month of birth, with bin size of 1 month each. Sample spans 2004-2007 of the EICS data and comprises mothers aged 18-40 in cohabitating or married relationships who have experienced a birth in the last year. Optimal Bandwidth chosen by the plug-in procedure suggested by Imbens and Kalyanaraman (2009).
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Table IV: Means and Difference-in-differences in Characteristics of Full EICS Sample

	Control 2002-2005	Control 2006-2010	Quebec 2002-2005	Quebec 2006-2010	Difference in Differences
Age of Mother	30.307	30.456	29.071	30.134	0.909***
Age of Spouse/Partner	32.732	32.851	31.900	32.586	0.552
Legally Married	0.862	0.846	0.366	0.389	0.040**
Immigrant	0.213	0.204	0.113	0.169	0.063
Family Size	3.784	3.830	3.711	3.760	0.006
Number of children aged 0-1	1.013	1.019	1.011	1.015	0.001
Number of children aged 1-5	0.526	0.582	0.497	0.528	-0.021
Number of children aged 6-17	0.243	0.251	0.205	0.249	0.036
Mother has high school degree or less	0.244	0.162	0.163	0.162	0.034***
Mother has some college	0.415	0.417	0.552	0.466	-0.087*
Mother has college degree	0.339	0.371	0.284	0.371	0.053
Father has high school degree or less	0.279	0.261	0.252	0.176	-0.060*
Father has some college	0.434	0.432	.512	0.515	-0.005
Father has college degree	0.285	0.306	0.235	0.308	0.054

*** p<0.01, ** p<0.05, * p<0.1

Notes: Sample spans 2002-2010 of the EICS data and comprises mothers aged 18-40 in cohabitating or married relationships who have experienced a birth in the last year. Difference-in-differences are identified across provinces and time.

Table V: Difference-in-Differences in Parents' Leave Outcomes

OUTCOMES:	(1) Fathers' Participation Rates	(2) Fathers' Leave Duration (Weeks)	(3) Mothers' Participation Rates	(4) Mothers' Leave Duration (Weeks)
Baseline for Quebec (2002-2005 Average)	0.213	2.011	0.725	43.140
I. Simple D-in-D Specification				
Quebec * Post-Reform	0.531***	3.225*	0.121*	2.845
	[0.00]	[0.08]	[0.07]	[0.25]
N	8907	7157	8907	6172
II. D-in-D Specification including Personal Covariates and Province & Year- Fixed Effects				
Quebec * Post-Reform	0.527***	3.241	0.125**	2.765
	[0.00]	[0.15]	[0.05]	[0.14]
N	8905	7156	8905	6441

*** p<0.01, ** p<0.05, * p<0.1, Heteroskedasticity-robust province-clustered p-values in parentheses

Notes: Table presents difference-in-difference estimates of parents' leave behavior between Quebec and Other Provinces before and after the introduction of QPIP in 2006. Sample spans 2002-2010 of the EICS data and comprises mothers aged 18-40 in cohabitating or married relationships who have experienced a birth in the last year. Heteroskedasticity-robust p-values, clustered at the province level and calculated using wild bootstrap procedures, are presented in brackets.

Table VI: Program Effect of QPIP on Joint Distribution of Parental Leave

	(1) Mother takes 0 months of leave	(2) Mother takes 1-5 months of leave	(3) Mother takes 6-9 months of leave	(4) Mother takes 10-11 months of leave	(5) Mother takes 12+ months of leave
(A)					
Father takes 0	0.002	-0.049	-0.070	-0.128**	-0.336***
weeks of leave	[0.95]	[0.12]	[0.71]	[0.03]	[0.00]
(B)					
Father takes 1-5	0.003***	-0.003***	0.089***	0.211***	0.258***
weeks of leave	[0.00]	[0.00]	[0.00]	[0.00]	[0.00]
(C)					
Father takes 6+	-0.001	-0.004	0.017**	0.021	-0.008
weeks of leave	[0.57]	[0.21]	[0.00]	[0.29]	[0.89]

*** p<0.01, ** p<0.05, * p<0.1, Heteroskedasticity-robust province-clustered p-values in parentheses

Notes: Table shows the difference-in-differences across provinces and time in the likelihood of various combinations of mothers and fathers' leave duration. Columns represent mothers' leave duration in months and rows represent fathers' duration in weeks; hence each cell represents a different outcome which is an indicator for a family choosing a particular combination of mothers' and fathers' leave durations. Sample spans 2002-2010 of the EICS data and comprises mothers aged 18-40 in cohabitating or married relationships who have experienced a birth in the last year. All regressions include controls for personal and household characteristics and province- and year- fixed effects.

Table VII: QPIP's Heterogeneous program effects by Household Income Levels

OUTCOMES:	(1) Fathers' Participation Rates	(2) Fathers' Leave Duration (Weeks)	(3) Mothers' Participation Rates	(4) Mothers' Leave Duration (Weeks)
I. Annual Household Income Threshold of 20,000CAD				
Baseline for Households above threshold	0.205	1.662	0.770	43.853
Baseline for Households below threshold	0.272	4.081	0.397	34.907
Quebec * Post-Reform	0.559***	3.770**	0.099**	2.327
	[0.00]	[0.05]	[0.04]	[0.19]
Household income below 20,000CAD*	-0.289***	-4.233***	0.242**	5.792
Quebec * Post-Reform	[0.00]	[0.00]	[0.04]	[0.00]
N	8905	7156	8905	5121
II. Annual Household Income Threshold of 40,000CAD				
Baseline for Households above threshold	0.191	1.596	0.758	44.85
Baseline for Households below threshold	0.234	2.347	0.696	41.580
Quebec * Post-Reform	0.633***	4.150***	0.080	-0.467
	[0.00]	[0.00]	[0.20]	[0.38]
Household income below 40,000CAD*	-0.233	-1.980	0.114**	7.721***
Quebec * Post-Reform	[0.26]	[0.32]	[0.02]	[0.00]
N	8905	7156	8905	5121

*** p<0.01, ** p<0.05, * p<0.1, Heteroskedasticity-robust province-clustered p-values in parentheses

Notes: Table reports the heterogeneous program effects of QPIP by exploiting variation in time, province and across income groups in difference-in-difference-in-difference regressions. Baselines provide 2002-2005 Averages in Quebec for observations in each group. Regressions control for Personal Covariates and Province & Year- Fixed Effects. Sample spans 2002-2010 of the EICS data and comprises mothers aged 18-40 in cohabitating or married relationships who have experienced a birth in the last year.

Table VIII: Mean Characteristics of GSS Data

	Parents with youngest child aged 1-3				Parents with youngest child aged 5-8				Difference-in-
	Control 2005	Control 2010	Quebec 2005	Quebec 2010	Control 2005	Control 2010	Quebec 2005	Quebec 2010	Difference-in Differences
Fathers' Characteristics (N=1606)									
Age	34.896	34.815	34.407	34.120	39.831	39.796	39.661	39.523	-0.102
Age of spouse	36.256	32.959	31.933	30.877	37.713	38.048	37.161	37.617	-1.103
Canadian Born	0.767	0.736	0.855	0.701	0.770	0.694	0.829	0.879	-0.248**
Born in Quebec	0.027	0.023	0.824	0.677	0.018	0.028	0.812	0.828	-0.150
Age of Youngest child	1.905	1.675	1.624	1.566	6.517	6.507	6.692	6.548	0.306
Children under 14	1.963	1.895	1.842	1.875	1.947	1.887	1.905	1.892	0.054
Children in Household	1.939	1.920	1.857	1.895	2.096	1.998	2.127	1.929	0.157
Legally Married	0.889	0.851	0.499	0.522	0.896	0.926	0.531	0.457	0.165
Household size	4.023	4.034	3.875	4.018	4.196	4.080	4.146	3.963	0.198
Mothers' Characteristics (N=1936)									
Age	32.465	32.600	31.748	31.111	36.945	38.748	37.042	38.485	-0.412
Age of Spouse	34.331	34.782	33.605	33.248	37.576	38.962	40.882	38.904	0.285
Canadian Born	0.728	0.674	0.786	0.814	0.729	0.692	0.803	0.754	0.094
Quebec Born	0.022	0.018	0.753	0.799	0.025	0.023	0.747	0.733	0.064
Age of Youngest child	1.841	1.606	1.853	1.534	6.400	6.483	6.610	6.232	0.369
Children under 14	1.828	1.929	1.736	1.777	1.952	2.012	2.009	1.941	0.068
Children in Household	1.827	1.924	1.730	1.800	2.160	2.075	2.086	2.038	-0.063
Legally Married	0.897	0.864	0.458	0.447	0.917	0.917	0.568	0.534	0.053
Household size	3.975	4.032	3.813	3.889	4.266	4.236	4.095	4.068	0.016

Notes: Data comprises a GSS sample of parents aged 18-50 who report being in a cohabitating or married relationship and whose youngest child is aged between one and eight years old. The last column presents difference-in-difference-in-difference in means across years, provinces and children's age groups.

Table IX: Baseline Sex Specialization in Quebec before the Reform

	Fathers	Mothers	$\frac{Fathers}{Mothers}$
Daily minutes in non-market work	158.70	312.83	0.51
Daily minutes in childcare	88.69	165.42	0.54
- Interactive childcare	42.56	61.96	0.69
- Routine childcare	37.77	87.72	0.43
- Travel & communication	8.36	15.73	0.53
Daily minutes in domestic Work	70.01	147.41	0.47
- Cooking	31.78	70.88	0.45
- Housekeeping	16.04	64.38	0.25
- Shopping	48.91	50.86	0.96
- Other chores	12.36	10.45	1.18
- Maintenance & repairs	9.80	1.69	5.79
Daily minutes at home	873.68	1,134.92	0.76
Daily minutes with family	409.07	509.32	0.80
Daily minutes in paid work	416.75	168.54	2.47
Daily minutes at workplace	314.44	113.14	2.77
Annual personal income	52,691	20,904	2.52
Hourly wages	29.450	22.512	1.31
Employed	0.836	0.572	1.46
- Full-time if employed	0.961	0.759	1.27
- Usual weekly hours if employed	44.025	31.109	1.42
- Weeks worked if employed	47.452	42.865	1.11

Notes: Table presents means from a sample of data from the 2005 GSS, comprising parents aged 18-50 who report being in a cohabitating or married relationship, living in Quebec, and having a youngest child aged 1-3 years old. Time spent at home and with family may include time spent sleeping.

Table X: Exposure to QPIP and Parents' Market Outcomes

OUTCOMES:	(1) Time in Paid Work	(2) Time at Workplace	(3) Annual Personal Income	(4) Employed	(5) Full-time Worker (if emp)	(6) Usual Weekly Hours (if emp)	(7) Weeks worked Last Year (if emp)
I. FATHERS							
Double-Differences using fathers with child aged 1-3							
Quebec * Post-reform	-43.296	-8.498	-2446.923	0.002	0.035	0.016	1.945***
	[0.16]	[0.70]	[0.17]	[0.90]	[0.12]	[0.90]	[0.00]
N	988	988	988	988	888	888	888
Triple-differences using all Fathers							
Child Under 3 *	-178.046***	-103.786**	-9539.84**	0.017	0.021	2.199	-2.03**
Quebec * Post	[0.00]	[0.02]	[0.04]	[0.44]	[0.40]	[0.19]	[0.04]
N	1596	1596	1596	1596	1280	1280	1280
II. MOTHERS							
Double-Differences using mothers with child aged 1-3							
Quebec * Post-reform	60.147***	52.287***	5819.181***	0.046*	0.054*	1.353*	-3.257***
	0.00	0.00	0.00	[0.09]	[0.07]	0.07	0.00]
N	1115	1115	1115	1115	692	692	692
Triple-differences using all Mothers							
Child Under 3 *	35.443	27.138*	10584.00***	-0.003	0.177**	4.295**	-3.031**
Quebec * Post	[0.23]	[0.08]	[0.00]	[0.92]	[0.01]	[0.03]	[0.04]
N	1939	1939	1939	1939	1278	1278	1278

*** p<0.01 ** p<0.05 * p<0.1, Robust province-clustered pvalues in parentheses

Notes: Data comprises a GSS sample of parents aged 18-50 who report being in a cohabitating or married relationship and whose youngest child is aged between one and eight years old. Results are from difference-in-difference-in-difference regressions exploiting variation across provinces, time and age of one's youngest child.

Table XI: Exposure to QPIP and Parents' Non-Market Outcomes

OUTCOMES:	(1) Total Time in Non-market Work	(2) Time in Domestic Work	(3) Time in Childcare	(4) Time at Home	(5) Time with Family
I. FATHERS					
Double-Differences using fathers with child aged 1-3					
Quebec * Post-reform	37.031***	15.312**	21.722***	36.371**	-29.811
	[0.00]	[0.01]	[0.00]	[0.02]	[0.22]
N	988	988	988	988	988
Triple-Differences using all fathers					
Child Under 3 *	43.404*	41.851**	1.542	67.797	29.103**
Quebec * Post	[0.07]	[0.02]	[0.90]	[0.11]	[0.03]
N	1596	1596	1596	1596	1596
II. MOTHERS					
Double-Differences using mothers with child aged 1-3					
Quebec * Post-reform	30.828**	-18.025**	48.853***	-30.342**	-11.237
	[0.01]	[0.04]	[0.00]	[0.01]	[0.68]
N	1115	1115	1115	1115	1115
Triple-Differences using all mothers					
Child Under 3 *	38.826***	-28.296***	67.123***	-33.936	-12.159
Quebec* Post	[0.00]	[0.00]	[0.00]	[0.16]	[0.17]
N	1939	1939	1939	1939	1939

*** p<0.01, ** p<0.05, * p<0.1, Heteroskedasticity-robust province-clustered p-values in parentheses

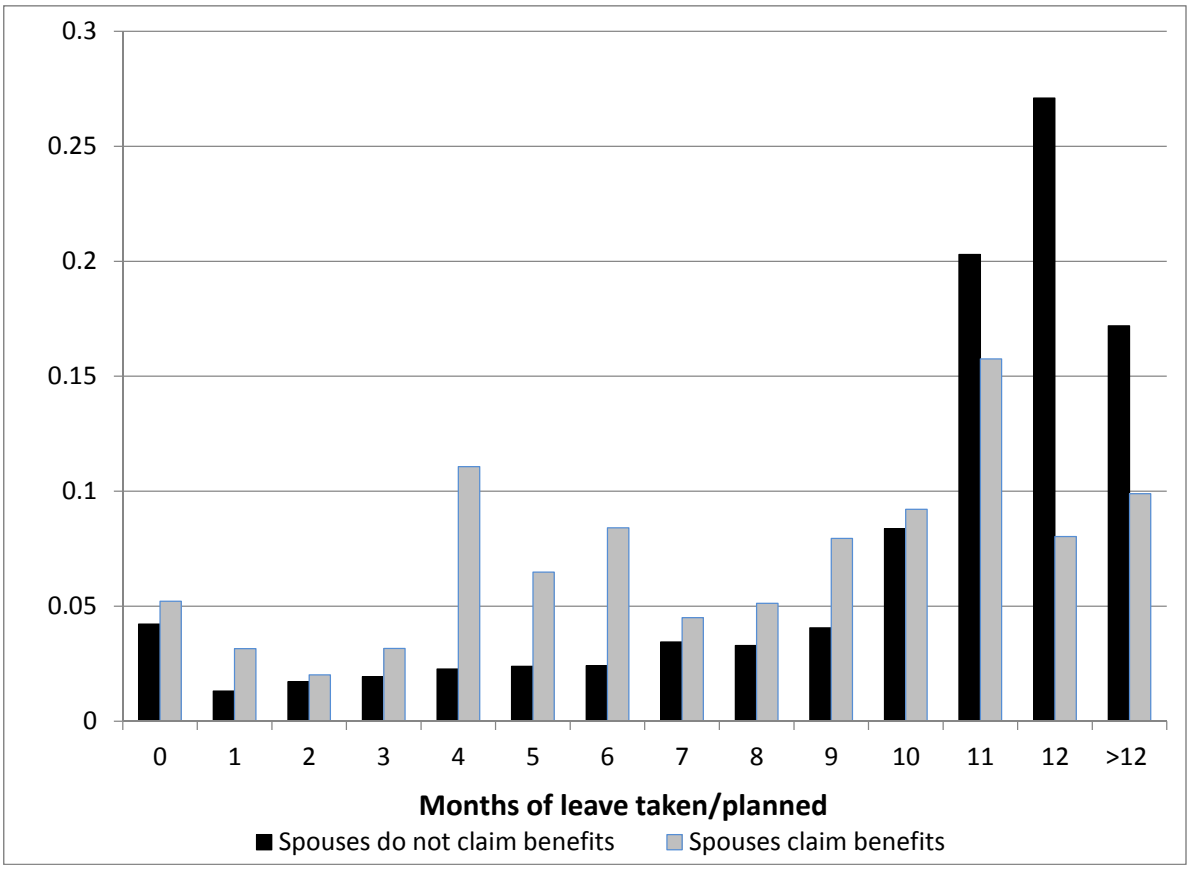

Figure I
PDF of Maternity Leave Duration in Quebec Pre-reform

Source: Constructed by author using Employment Insurance Coverage Survey data from 2002-2005

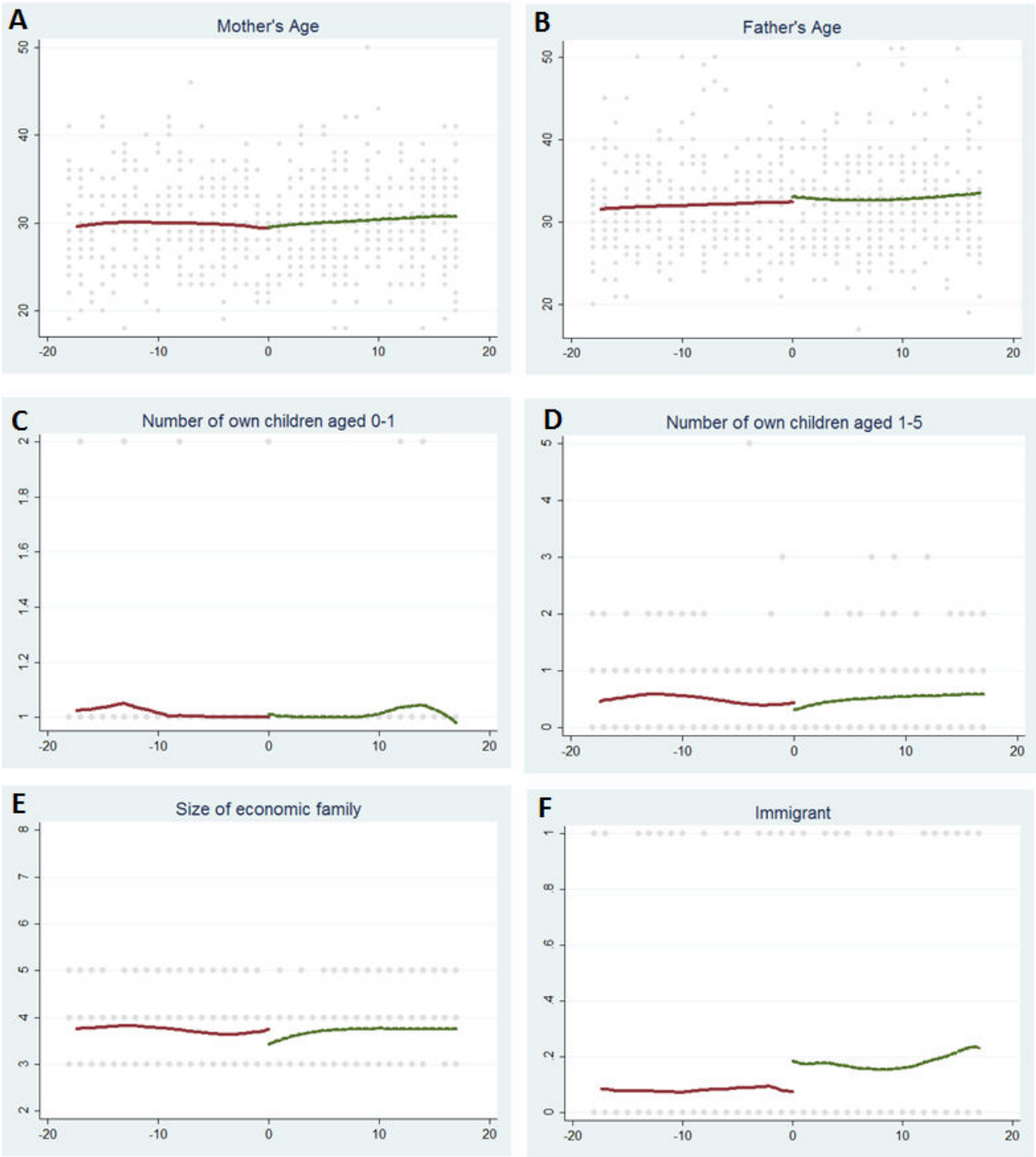

Figure II
Discontinuities in household characteristics in EICS data

Source: Graphs created from non-parametric local linear regressions using EICS data for Quebec to identify jumps in mothers' leave duration at the cutoff of January 2006. Corresponding Regression results in Table 2.

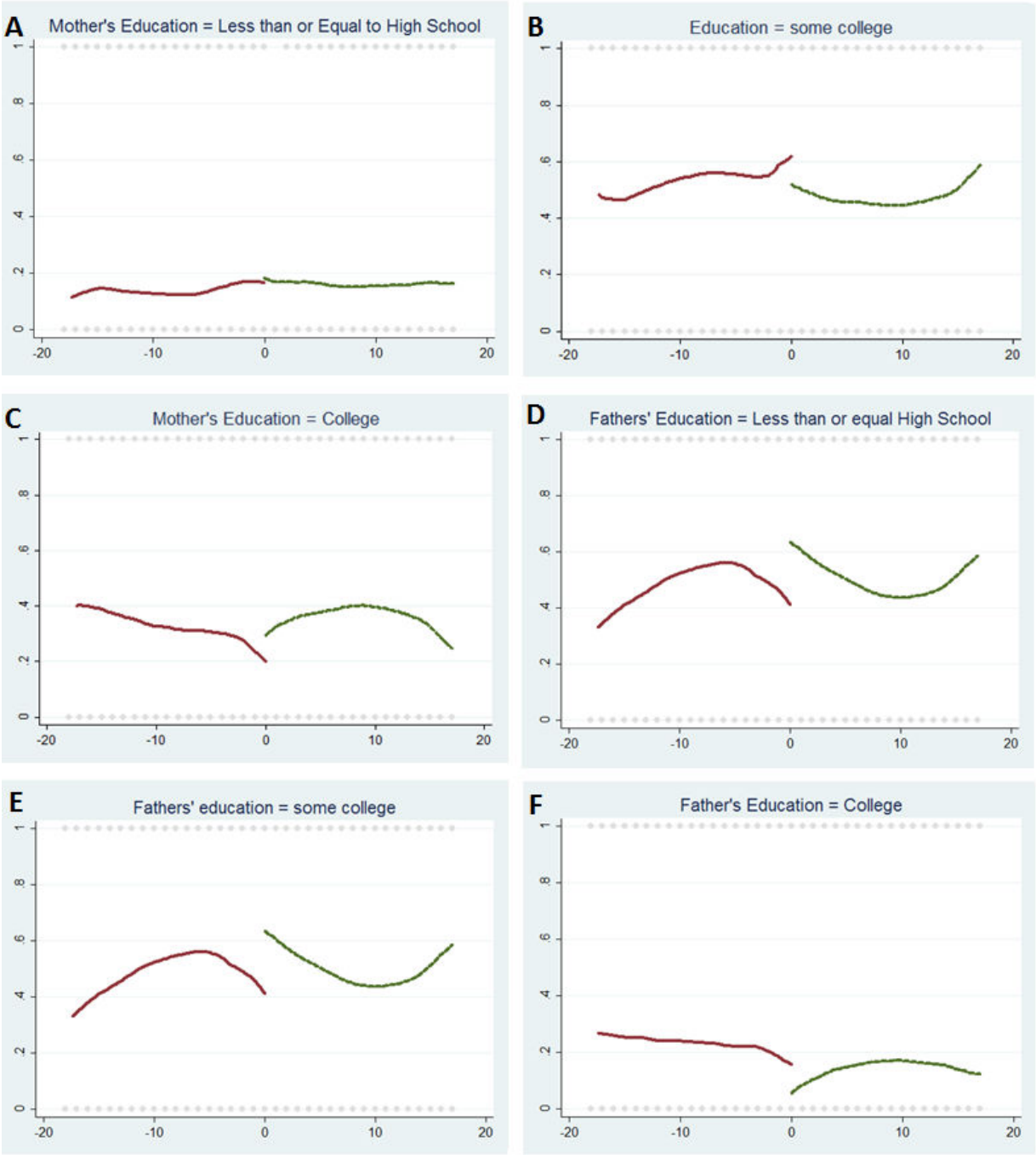

Figure III
Discontinuities in educational characteristics in EICS data

Source: Graphs created from non-parametric local linear regressions using EICS data for Quebec to identify jumps in mothers' leave duration at the cutoff of January 2006. Corresponding Regression results in Table 2.

Figure IV
Discontinuities in Parents' Leave Outcomes

Source: Graphs created from non-parametric local linear regressions using EICS data for Quebec to identify jumps in mothers' leave duration at the cutoff of January 2006. Corresponding Regression results in Table 3

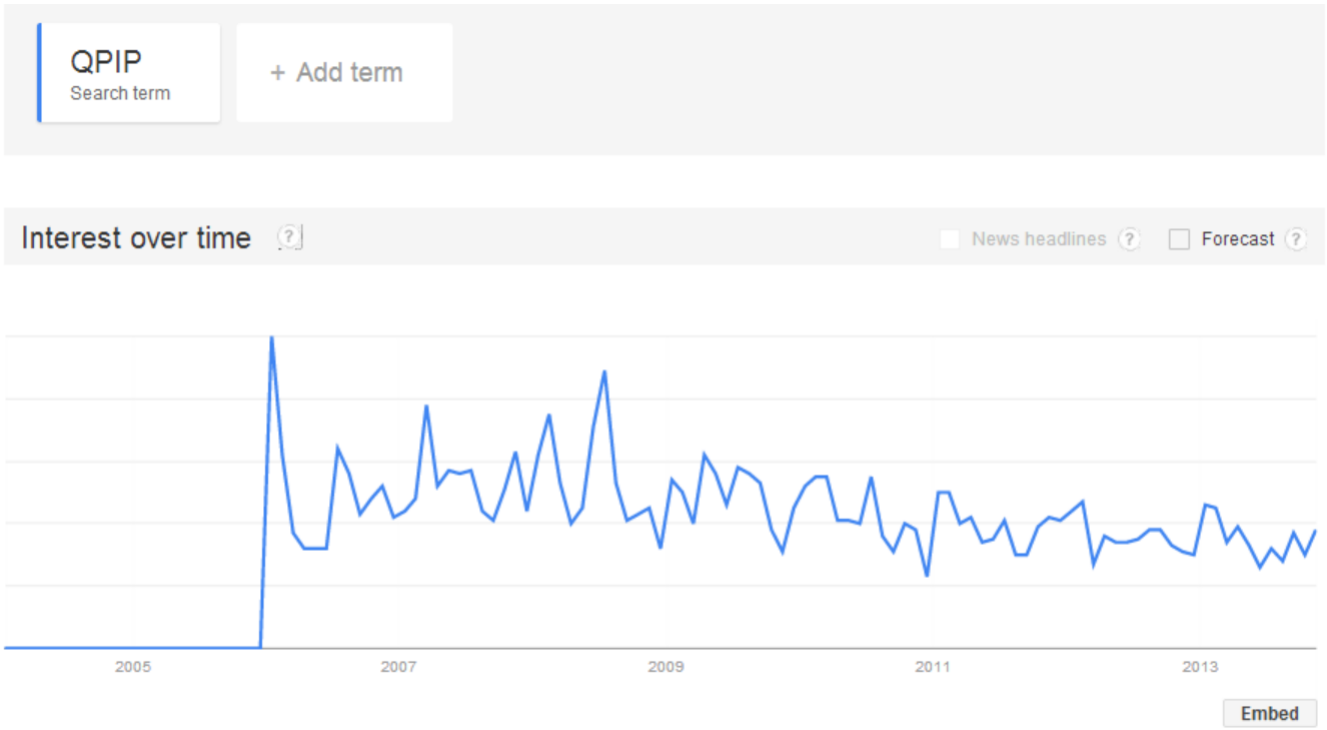

Figure V
Google Trends in Searches for the word 'QPIP'

Source: Graph collected by author from using Google Trends Search. Similar results were obtained when using terms such as "Quebec parental insurance program" or "Regime quebecois de l'assurance parentale".

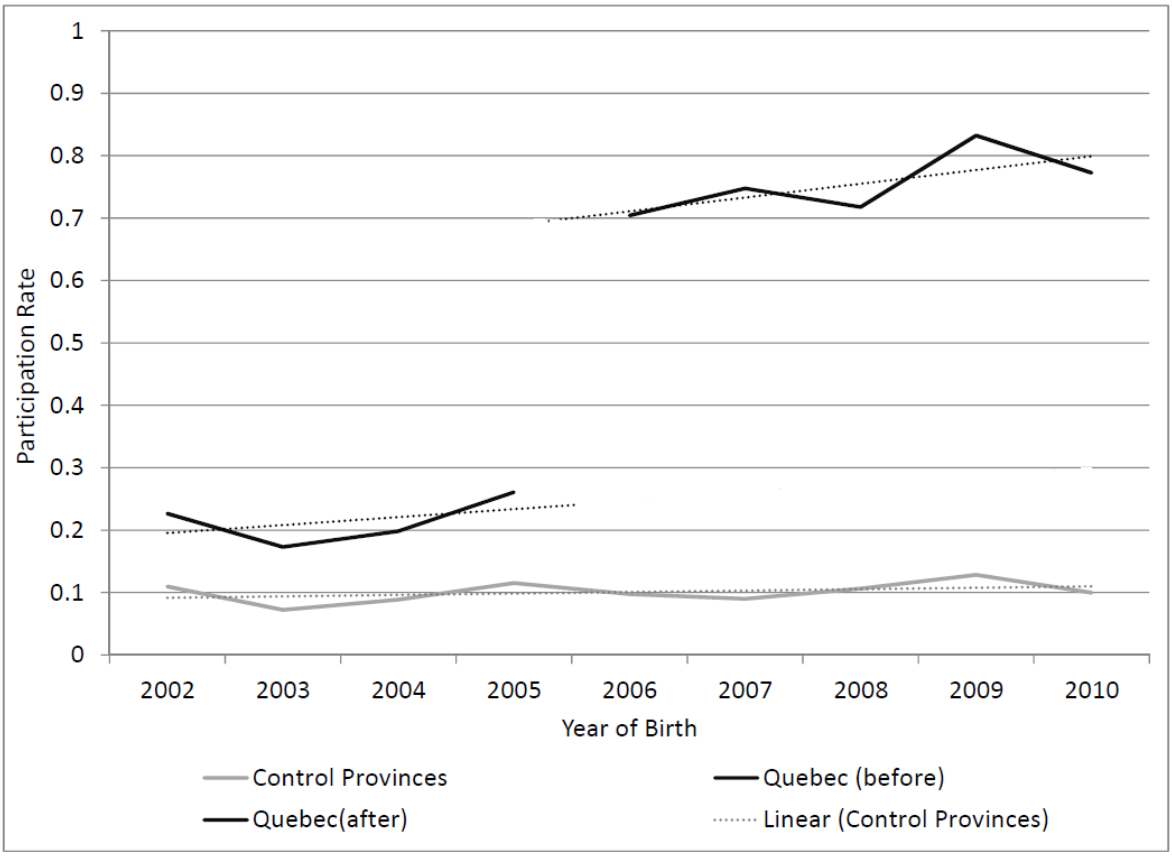

Figure VI
Parallel Trends in Fathers' Leave Participation

Source: Constructed by author using Employment Insurance Coverage Survey data from 2002-2010

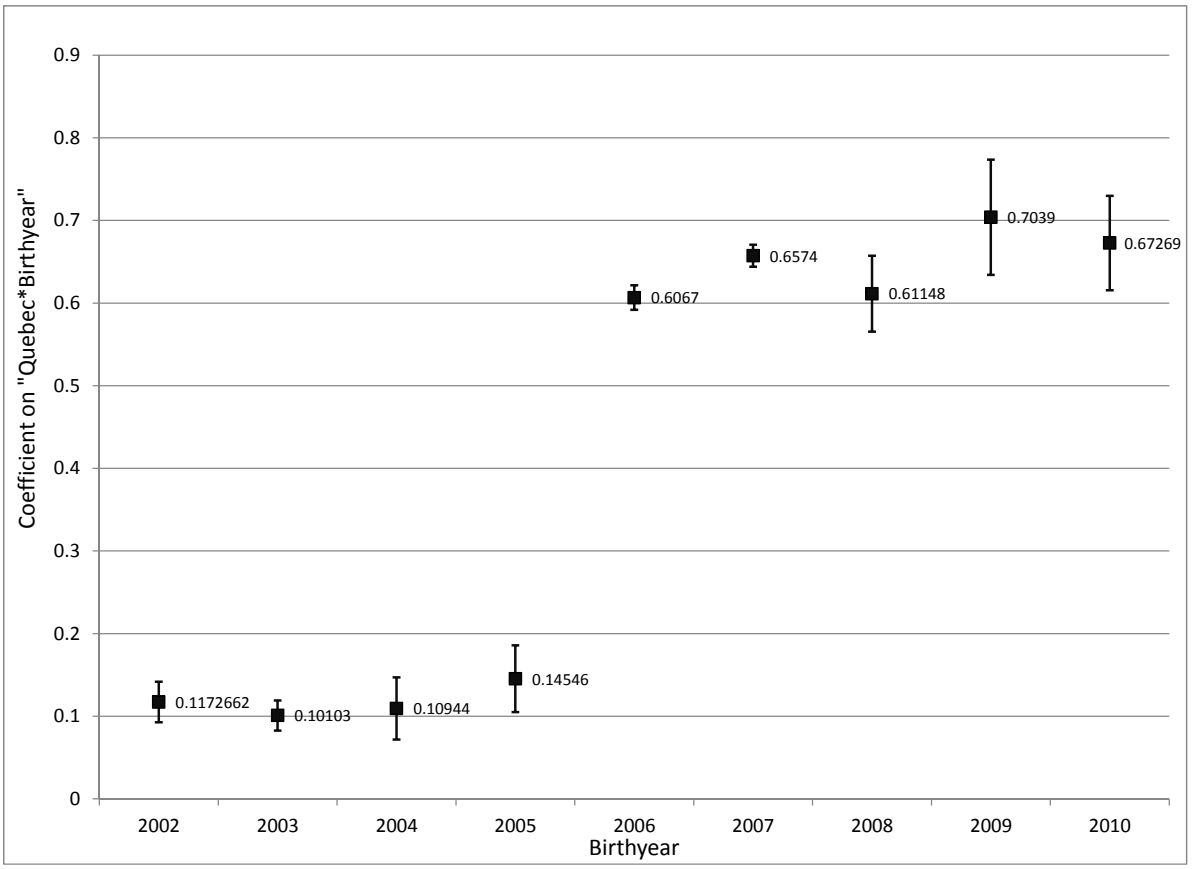

Figure VII
Event Study of Fathers' Leave Participation Rates

Source: Constructed from an event study conducted by running a regression of the outcome on indicators for each year as well as interaction terms between an indicator for being in Quebec and indicators for each year. Graph plots coefficients from the interaction terms of Quebec and indicators for every year, and thus presents the difference between the average father in Quebec and that in other provinces in every given year between 2002-2010. Vertical lines in black show 95% confidence interval around the coefficient.

Appendix A: Robustness Checks

Table A.I: Parametric RD Analyses of Quebec[25]

	Fathers' Participation Rates	Fathers' Leave Duration (Weeks)	Mothers' Participation Rates	Mothers' Leave Duration (Weeks)
I. Bandwidth = 24 months				
RD Estimate	0.521***	3.611**	-0.022	-0.321
	[0.00]	[0.03]	[0.78]	[0.95]
N	806	786	806	617
II. Bandwidth = 18 months				
RD Estimate	0.500***	3.339	0.072	1.757
	[0.00]	[0.11]	[0.36]	[0.72]
N	606	590	606	464
III. Bandwidth = 12 months				
RD Estimate	0.601***	4.701**	0.245	4.611
	[0.00]	[0.01]	[0.62]	[0.53]
N	396	391	396	305

*p<0.10 **p<0.05 ***p<0.01, Heteroskedasticity-robust p-values in brackets

Notes: Table present results from Parametric RD regressions on treatment province of Quebec to detect discontinuities in outcomes at point of cutoff. Sample comprises births in months surrounding the reform, with the window varying from 12 to 18 to 24 months on either side of 1st January 2006. Functional form for each regession was selected using the F-test method suggested by Lee and Lemieux (2010). Regressions include controls for Personal covariates e.g. age and education of mother and spouse as well as household size, religion and number of children in various age groups. Errors are allowed to cluster around the month of birth. Robust p-values presented in square brackets.

Table A.II: Non-Parametric RD Analysis of Quebec, Trimming around cutoff

	(1) Fathers' Participation Rates	(2) Fathers' Leave Duration (Weeks)	(3) Mothers' Participation Rates	(4) Mothers' Leave Duration (Weeks)
Jump at Cutoff	0.531***	1.855	0.194*	4.888
in Quebec	[0.00]	[0.21]	[0.08]	[0.12]
Bandwidth (months)	9.104	19.899	8.884	24.408

*p<0.10 **p<0.05 ***p<0.01, Heteroskedasticity-robust clustered p-values in brackets

Notes: Table presents results from RD regressions to detect discontinuities in outcomes at the point of cutoff. Regressions conducted non-parametrically using local linear regression methods, and optimal bandwidth selected using the plug-in procedure suggested by imbens2009. Births from January 2006 and December 2005 are excluded from this sample. Regressions include controls for Personal covariates e.g. age and education of mother and spouse as well as household size, religion and number of children in various age groups. Errors are allowed to cluster around the month of birth. Robust p-values presented in square brackets.

Table A.III: Difference in Differences in Conditional Leave Duration

	(1) Fathers' Leave Duration (Weeks)	(2) Mothers' Leave Duration (Weeks)
Baseline for Quebec (2002-2005 Average)	10.856	46.246
I. Simple D-in-D Specification		
Quebec * Post-Reform	-4.337	0.966
	[0.85]	[0.86]
N	6,236	5,789
II. D-in-D Specification including Personal Covariates and Province & Year- Fixed Effects		
Quebec * Post-Reform	-4.466	0.868
	[0.86]	[0.84]
N	6,236	5,789

*p<0.10 **p<0.05 ***p<0.01, Heteroskedasticity-robust province-clustered p-values in brackets

Notes: Outcome is measured as weeks of leave planned or taken by the parent conditional on them reporting participation. Sample spans 2002-2010 of the EICS data and comprises mothers aged 18-40 in cohabitating or married relationships who have experienced a birth in the last year. Regressions control for personal and household characteristics and province and year fixed effects.

Table A.IV: Summary Test for Sample Composition Bias

	(1) Predicted Fathers' Participation Rates	(2) Predicted Fathers' Leave Duration (Weeks)	(3) Predicted Mothers' Participation Rates	(4) Predicted Mothers' Leave Duration (Weeks)
Quebec * Post-Reform	-0.004	-.051	-0.025*	-0.755***
	[0.45]	[0.79]	[0.07]	[0.00]
N	8,096	6,236	8,096	5,789

*p<0.10 **p<0.05 ***p<0.01, Heteroskedasticity-robust province-clustered p-values in brackets

Notes: Table presents difference-in-difference estimates of parents' predicted leave behavior between Quebec and Other Provinces before and after the introduction of QPIP in 2006. The dependent variable is a predicted outcome obtained by regressing the participation rate or leave duration on a host of personal and household characteristics on the pre-treatment sample i.e. mothers in Quebec before 2006. Sample spans 2002-2010 of the EICS data and comprises non-immigrant mothers aged 18-40 in cohabitating or married relationships who have experienced a birth in the last year. Set of personal and household characteristics include age of respondent and spouse, legal marital status, indicator for immigrant, household size, number of children aged 0-1 and 1-5, and various education levels of respondent and spouse. Province and year- fixed effects are controlled for.

Table A.V: Difference-in-Differences in leave outcomes, using sub-samples of parents

	(1) Fathers' Participation Rates	(2) Fathers' Leave Duration (Weeks)	(3) Mothers' Participation Rates	(4) Mothers' Leave Duration (Weeks)
I. Non-Immigrant Mothers				
Quebec * Post-Reform	0.548***	3.240**	0.125**	2.203
	[0.00]	[0.01]	[0.01]	[0.14]
N	7,753	6,236	7,753	5,537
II. Young Mothers aged 18-35				
Quebec * Post-Reform	0.524***	3.040	0.125**	2.488**
	[0.00]	[0.21]	[0.01]	[0.05]
N	7,790	6,245	7,790	5,383
III. Young Mothers aged 18-30				
Quebec * Post-Reform	0.534***	3.201	0.137*	2.297*
	[0.00]	[0.21]	[0.08]	[0.07]
N	4,888	3,876	4,888	3,274
IV. Mothers who are not legally married				
Quebec * Post-Reform	0.528***	2.581**	0.091	3.139
	[0.00]	[0.04]	[0.35]	[0.10]
N	2,387	1,932	2,387	1,726
V. Mothers who have more than high school education				
Quebec * Post-Reform	0.556***	3.289*	0.126***	2.663**
	[0.00]	[0.09]	[0.00]	[0.03]
N	2,048	1,597	2,048	1,035
VI. *Mothers with spouses who have high school education or less				
Quebec * Post-Reform	0.486***	3.95**	0.080	-2.048
	[0.00]	[0.03]	[0.97]	[0.81]
N	2,179	1,799	2,179	1,565

*p<0.10 **p<0.05 ***p<0.01, Heteroskedasticity-robust province-clustered p-values in brackets

Notes: Sample spans 2002-2010 of the EICS data and comprises mothers aged 18-40 in cohabitating or married relationships who have experienced a birth in the last year. Regressions constrol for personal and household characteristics and province and year- fixed effects. Subsamples were selected using information from Table 4 to determine the groups that appear to have become less prominent over time in the EICS Sample in Quebec over time.

Table A.VI: Difference-in-differences in fathers' leave participation, 2004-2010

	(1) Fathers' Participation Rates	(2) Fathers' Participation Rates
Quebec * Post-Reform	0.521***	0.521***
	[0.00]	[0.00]
Includes Personal Covariates and Province and Year FEs	No	Yes
N	7,288	7,287

*p<0.10 **p<0.05 ***p<0.01, Heteroskedasticity-robust province-clustered p-values in brackets

Notes: Sample is restricted to the years 2004-2010 of the EICS data to ensure comparability with data on fathers' leave duration which was only collected in those years. Sample comprises mothers aged 18-40 in cohabitating or married relationships who have experienced a birth in the last year. Regressions control for set of personal and household characteristics and province- and year- fixed effects.

Table A.VII: Exposure to QPIP and Parents' Hourly Wages

	(1)
OUTCOMES:	Log (Hourly) Wages
A. FATHERS :	
Child Under 3 * Quebec	-0.326***
* Post-Reform	[0.00]
N	1089
B. MOTHERS:	
Child Under 3 * Quebec	0.537***
* Post-Reform	[0.00]
N	855

Notes: Data comprises a GSS sample of parents aged 18-50 who report being in a cohabitating or married relationship and whose youngest child is aged between one and eight years old. Results are from difference-in-difference-in-difference regressions exploiting variation across provinces, time and age of one's youngest child. I restrict the sample to parents whose hourly wages fall in the range of 5.33CAD-336CAD in 2005 dollars.

Table A.VIII: Difference-in-Differences in Placebo Parents' Long-term Market Outcomes

OUTCOMES:	(1) Time in Paid Work	(2) Time at Workplace	(3) Annual Personal Income	(4) Employed	(5) Full-time Worker (if employed)	(6) Usual Weekly Hours (if employed)	(7) Weeks worked Last Year (if employed)	(8) Log Hourly Wages
A. FATHERS								
Quebec * Post-reform	144.370***	101.685**	3777.617	-0.016	0.0131	-1.864	3.759***	0.125*
	[0.00]	[0.01]	[0.14]	[0.48]	[0.38]	[0.20]	[0.06]	0.873
N	1115	1115	897	1115	692	694	690	424
B. MOTHERS								
Quebec * Post-Reform	31.88	29.859**	-3951.298**	0.056**	-0.116**	-2.545	-0.217	-0.10
	[0.12]	[0.02]	[0.01]	[0.03]	[0.03]	[0.11]	[0.76]	[0.11]
N	824	824	660	824	586	583	584	411

*** p<0.01, ** p<0.05, * p<0.1, Robust province-clustered p-values in parentheses

Notes: Table presents results from double-difference regressions exploiting variation across provinces and time, for potentially exposed parents (those with youngest child aged 1-3) and placebo parents (those with youngest child aged 5-8) separately. Sample restricted to mothers and fathers aged 18-50 in married or cohabitating relationships whose youngest child is aged 1-8 years old.

Table A.IX: Difference-in-Differences in Placebo Parents' Long-term Non-market outcomes

OUTCOMES:	(1) Total Time in Non-market Work	(2) Time in Domestic Work	(3) Time in Childcare	(4) Total Time Spent at Home	(5) Time with Family
A. FATHERS					
Quebec * Post-reform	-6.116	-31.587**	25.471***	-35.890	-63.756**
	[0.62]	[0.03]	[0.00]	[0.11]	[0.01]
N	1115	1115	1115	1115	1115
B. MOTHERS					
Quebec * Post-Reform	-10.100	8.811	-18.911**	2.584	-5.95
	[0.31]	[0.26]	[0.04]	[0.88]	[0.65]
N	824	824	824	824	824

*** p<0.01, ** p<0.05, * p<0.1, Robust province-clustered p-values in parentheses

Notes: Table presents results from double-difference regressions exploiting variation across provinces and time, for potentially exposed parents (those with youngest child aged 1-3) and placebo parents (those with youngest child aged 5-8) separately. Sample restricted to mothers and fathers aged 18-50 in married or cohabitating relationships whose youngest child is aged 1-8 years old.

Appendix B: Definitions of Variables Constructed by Author

Annual Personal Income is measured by annual income for the last year reported by the respondent. Income reported in survey year 2010 is adjusted for inflation to 2005 Canadian dollars by dividing by a factor of 1.07. This inflation factor of 1.07 is obtained using the inflation calculator provided by the Bank of Canada at http://www.bankofcanada.ca/rates/related/inflation-calculator.

Hourly wages are only calculated for respondents who report their main source of income as wages and salaries, and who have non-missing values for income, annual weeks worked last year and usual weekly hours. Hourly wages are calculated by dividing inflation-adjusted personal income for the last year by usual hours worked per week multiplied by weeks worked last year. One important issue is that sample sizes are smaller for this outcome, since we cannot calculate hourly wages for anyone with a missing value of personal income, annual weeks worked last year or usual hours worked. Another issue is that although the sample is restricted to parents who report that their main source of income is wages and salaries, the annual income amount reported may still include funds from other sources, making the issue of implausible or unlikely hourly wages particularly important. A few individuals do report implausibly high or low earnings relative to their hours of work, and the inclusion of these observations impacts the estimated mean and variance of wages. I follow the literature by trimming the data to remove extreme values and checking the sensitivity of my estimates to various wage restrictions. For example, in one set of regressions, I exclude observations with hourly wages below 2.66CAD in 2005 dollars (roughly half the minimum wage in most provinces) or above 268CAD in 2005 dollars - this corresponds to the $1-$100 (in 1979 dollars) range used by e.g. Card and DiNardo (2002) and Lemieux (2002). I also try setting the minimum to 5.33CAD (roughly the minimum wage) and/or the maximum to 336CAD (approximately $150 in 1979 dollars). My results for both fathers and mothers' log wages are highly consistent across all these specifications. The results shown in Appendix Table A.VII correspond to regressions using the wage range of 5.33CAD-336CAD in 2005 dollars.

Full-time Employment comprises a reported observation of 35 or more hours usual weekly hours worked. This choice of cutoff follows a large economic literature on the subject.